BODO REMEDIATION ANALYTICS-2024 LAND USE/LAND COVER STUDIES

Bodo Remediation Area of Interest
(16/22 January 2024)

Introduction

The ongoing Bodo project involves Shoreline Clean-up and Assessment Technique (SCAT) activities including onsite verification of remediation works, field sampling/laboratory analyses and mangrove revegetation activities, all running concurrently. To compliment field observations, it was decided that a time lapse analysis of satellite imagery should also be applied for the monitoring of mangrove recovery within the area of interest.

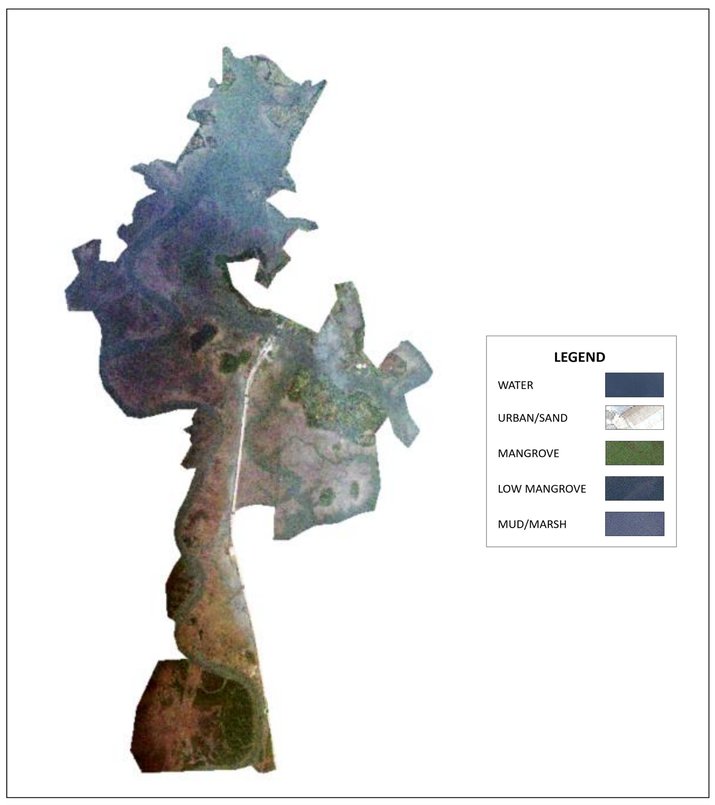
This report contains the statistical analysis result of the land use/land cover classification of the satellite imagery over the area of interest acquired on 16th/22nd January 2024 to highlight the current area extent of the vegetation cover as at the time of satellite data capture.

Methodology

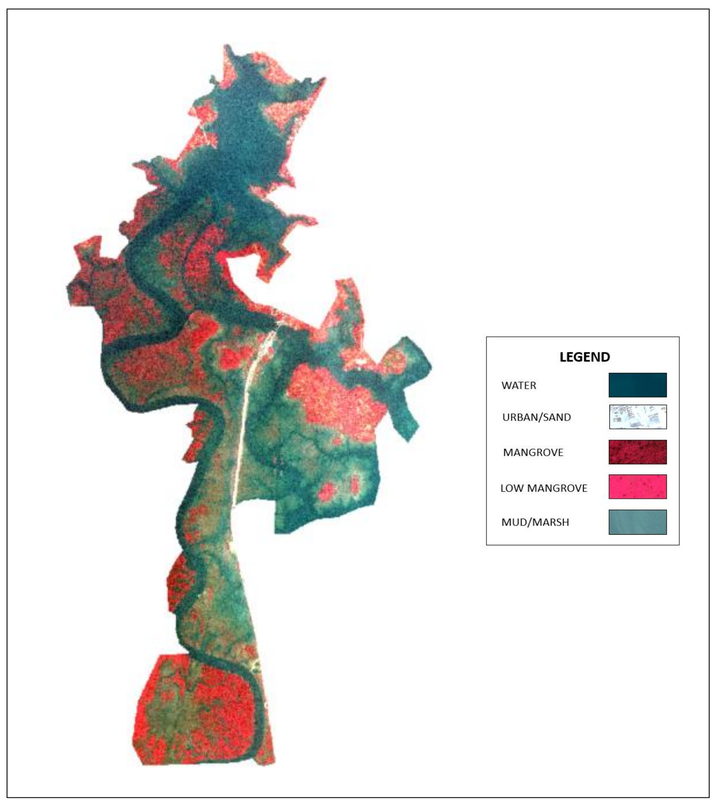
The satellite data was first corrected for the attenuating effect of atmospheric moisture, dust particles and fine clouds (so called atmospheric correction) and referencing/positioning errors. A 'False Colour' (FC) image was then generated from the corrected data i.e., the near-infrared reflection intensity of the image was visualized in red to highlight the chlorophyll cells in plant leaves which strongly absorb red EM radiation, but at the same time strongly reflect near-infrared wavelengths. The reddish hues gave an indication of vegetation of varying intensity and coverage, whereas blue, grey and white hues indicate water, mud/marsh and built-up areas (both categorize as Others in the 2024 classification result).

The satellite image data (16th/22nd January 2024) was first merged, georeferenced with the 2022 satellite image and classified using the support vector machine classification method from the classification wizard tool in ArcGIS Pro to generate land use and land cover class distribution for the area of interest. The baseline information for each class was extracted from the False Colour Composites (FC) combination of the RGB raster data and a statistical analysis of the identified feature classes was done to quantify the area of each land use/land cover class.

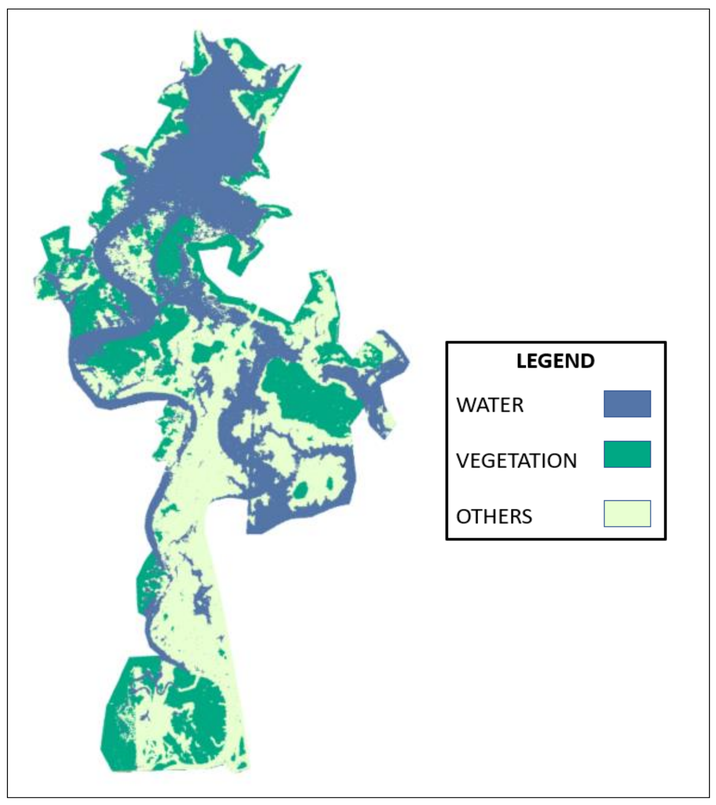
Results



True Colour Composite 16/22 January 2024

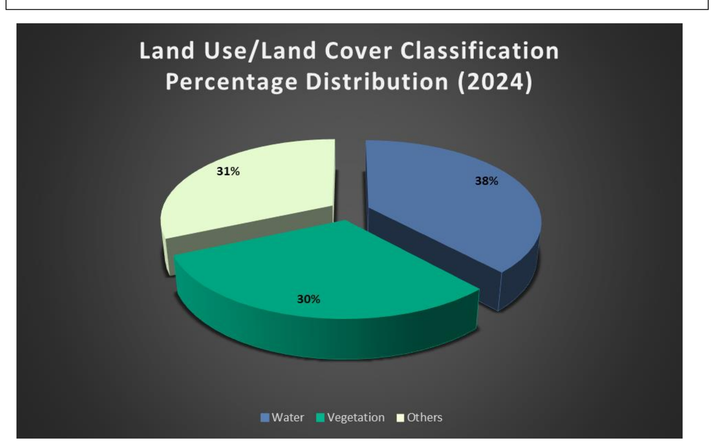


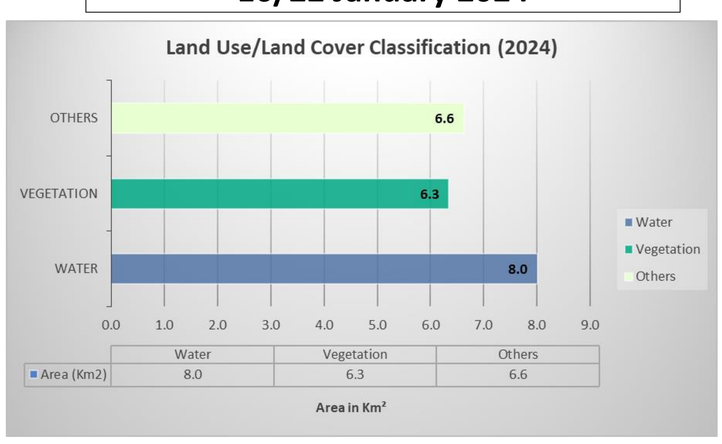
False Colour Composite 16/22 January 2024



Land Use/Land Cover Classification
16/22 January 2024

Epoch: 2024		
Land Use/Land Cover Types	**Area (Km2)**	**Percentage (%)**
Water	8.0	38%
Vegetation	6.3	30%
Others	6.6	31%
TOTAL	20.98	100%





Summary

The result of the statistical analysis for the 2024 classification study shows a land use/land cover percentage area distribution for **Water features** at **38%**, **Vegetation features** at **30%** and features categorize as **Others** at **31%**.

Water: The water class increased from 5.09km2 in 2022 to 8.0km2 in 2024. This increase may be attributed to changes in the water levels due to hide tide evident on the 2024 optical image used for the study.

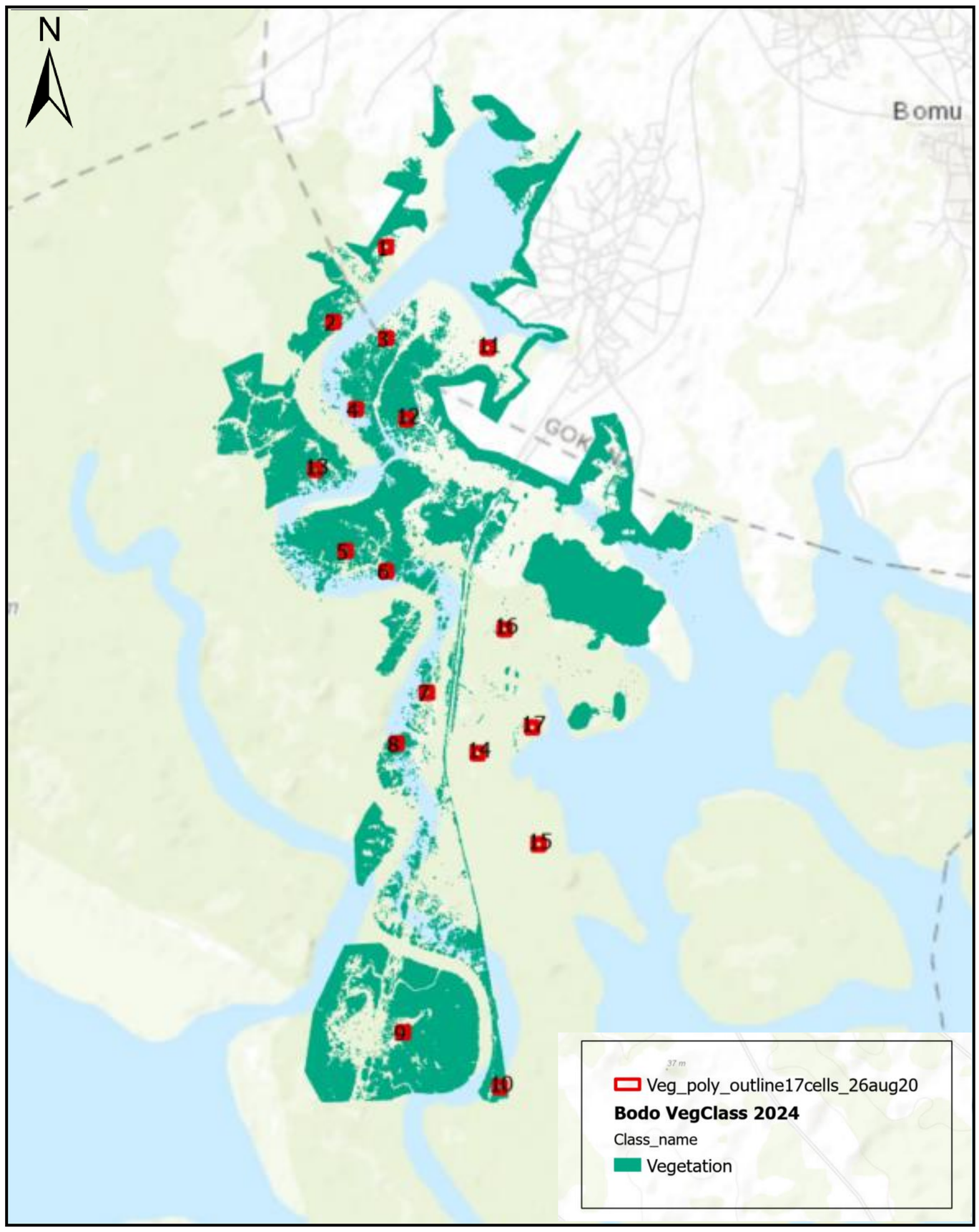
Vegetation: The vegetation class increased from 5.93km2 in 2022 to 6.3km2 in 2024. This may be attributed to progress of the ongoing Bodo mangrove revegetation exercise undertaken by the Ogoni restoration team/JV partners across the entire Bodo AOI. From the analysis, we propose that Cells 1, 7, 11, 16 and 17 be given adequate attention during the phase of the mangrove revegetation exercise as these Cells haven't experienced any change in the distribution of vegetation pattern and coverage since the last 2022 study year.

Others: The others class decreased from 9.96km2 in 2022 to 6.6km2 in 2024. The reason for the decrease is probably attributed to changes in the water levels due to hide tide forming large estuaries and extended riverbank covering most of the mud flats area. This is evident when comparing the classified result for both the 2022 and 2024 study year.

The result of this analysis fed into a more robust analysis of all available satellite imagery over the Bodo remediation area of interest till date to have a clear view of the time lapse change in mangrove distribution over time. A further change detection analysis was done using the Veg_poly_outline17cells_26aug20 and the extracted 2022/2024 vegetation classified data to highlight the evidence of vegetation recovery within the area of interest. (See page 2 of this report).

PREPARED BY: Amaechi Ojeh(IUV/C/NGI) **Geomatics Project Team: Adeoye Ogunleye (IUV/C/NGI), Amaechi Ojeh (IUV/C/NGI), Tunde Shebolatan (UPV/C/NGI)** **DATE: 23 May 2024**

CHANGE ANALYSIS FOR MANGROVE RECOVERY WITHIN BODO AOI (2022 & 2024)



Clip showing extracted vegetation class and sample cell locations for Bodo Remediation Area of Interest

Clip Description: ***Overview Map of extracted vegetation class from the 2024 classified Bodo imagery overlayed with Mangr_2020_veg_polys_17cells_31aug20 and 100 x 100m2 Veg_poly_outline17cells_26aug20 vector data***

Methodology:

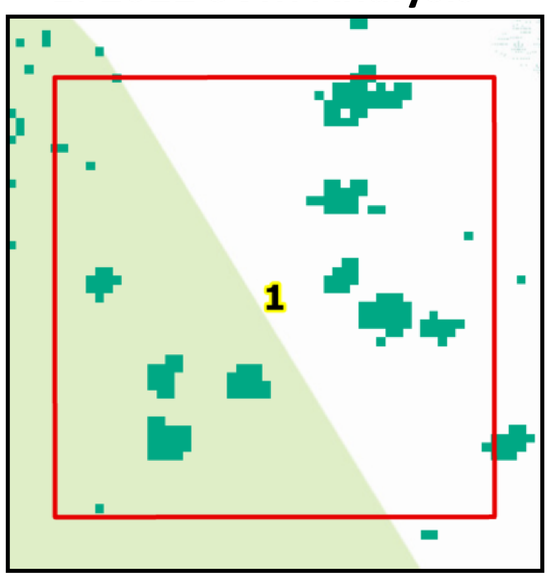
A change detection analysis was done across the eight research epochs (2013–2024) to identify the evidence of vegetation recovery within the area of interest and to determine the differences in the status of land use and land cover within the Bodo study area.(**See pages 14 – 15**).

Furthermore, in order to map out the different variations and estimate the progress of the mangrove regrowth within the Bodo AOI, a total number of 17 cell sites were identified to help measure each mangrove presence visually. These measurements were then used as a basis for comparing the image processing results for those areas. These 17 cell sample locations were selected to have various levels of mangrove coverage and were roughly distributed within the area of interest with a focus near the Bodo City. This principle was applied to the classification analyses over the Bodo remediation site (2022-2024) to visually interpret and analyze the amount of vegetation regrowth within each vegetation sample cell.

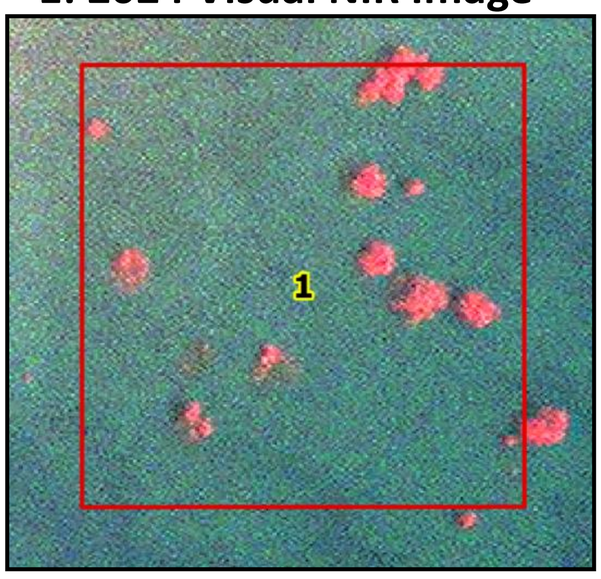
Below are the interpretation and classification results for the 2022 SVM Analysis and the 2024 SVM Analysis using the 17cells vegetation polygon outline data integrated with the 2024 Bodo NIR raster data. Each cells have been categories as **1 - 17**. **Cells 1 – 13 and 16 – 17** are within the Bodo cleanup AOI while **Cells 14 and 15** are within the Ataba AOI. Margin of Error between the 2022 SVM data and the 2024 SVM data Classification is + or – 5m.

Comparing Classification Results for 2022 SVM Analysis and 2024 SVM Analysis

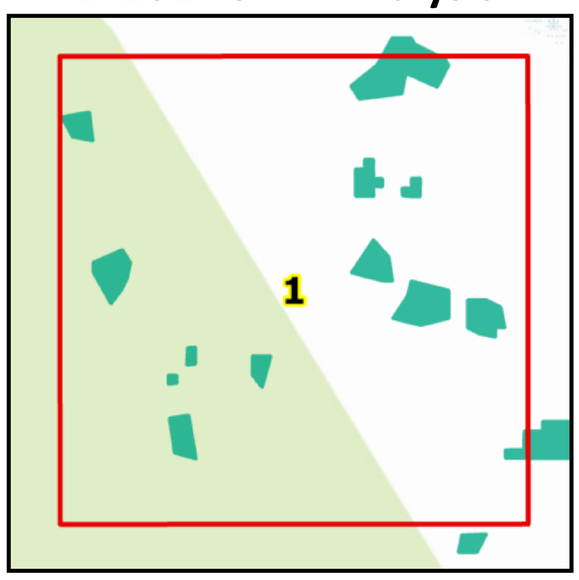
1: 2022 SVM Analysis



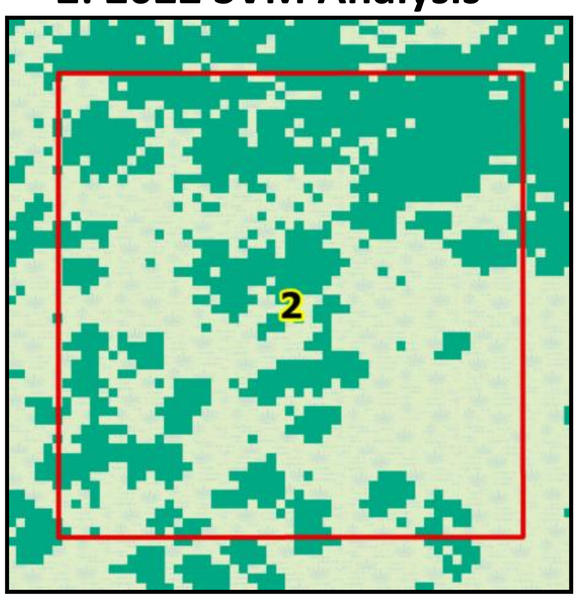
1: 2024 Visual NIR Image



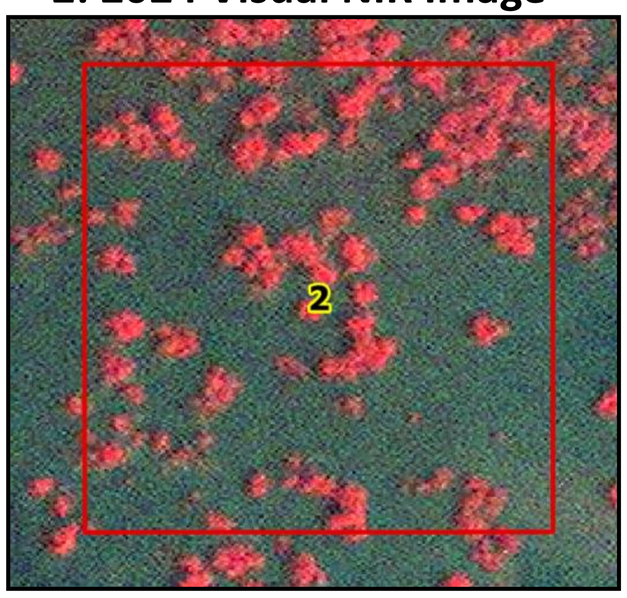
1: 2024 SVM Analysis



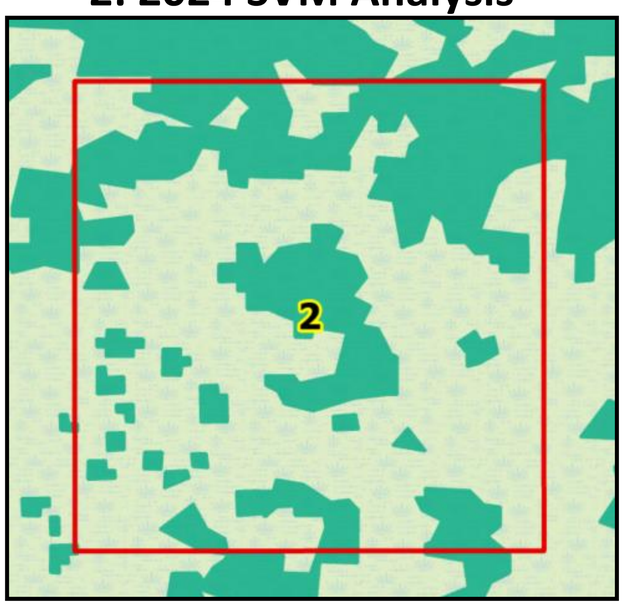
2: 2022 SVM Analysis



2: 2024 Visual NIR Image



2: 2024 SVM Analysis



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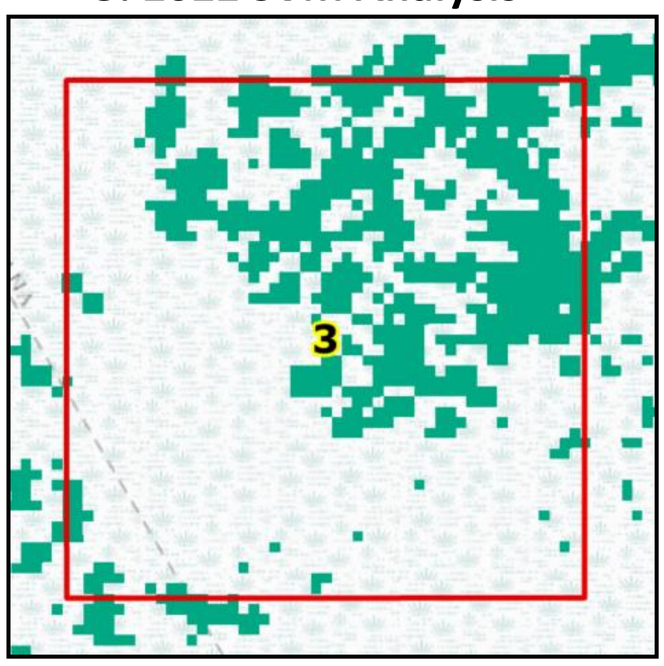
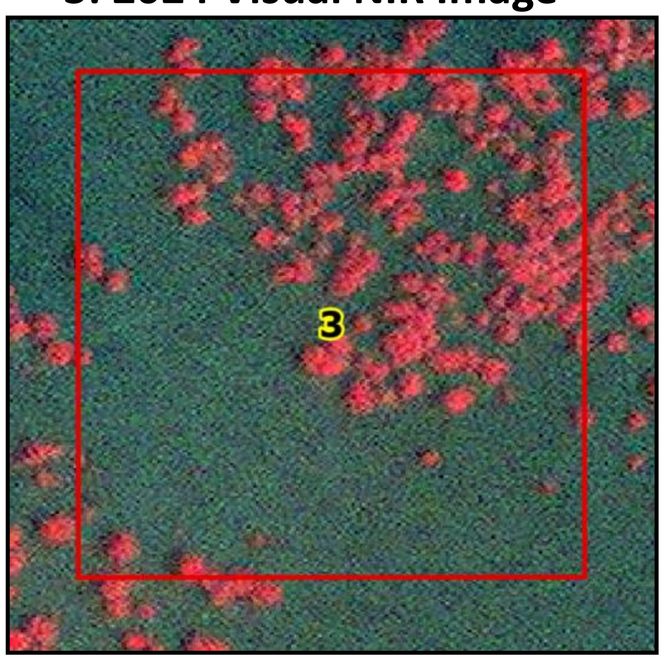
CHANGE ANALYSIS FOR MANGROVE RECOVERY WITHIN BODO AOI (2022 & 2024)

Comparing Classification Results for 2022 SVM Analysis and 2024 SVM Analysis

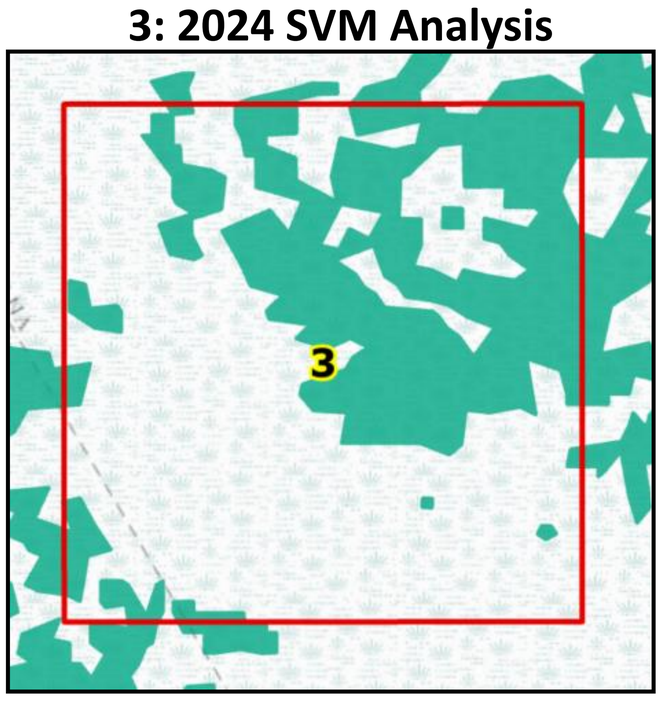
3: 2022 SVM Analysis



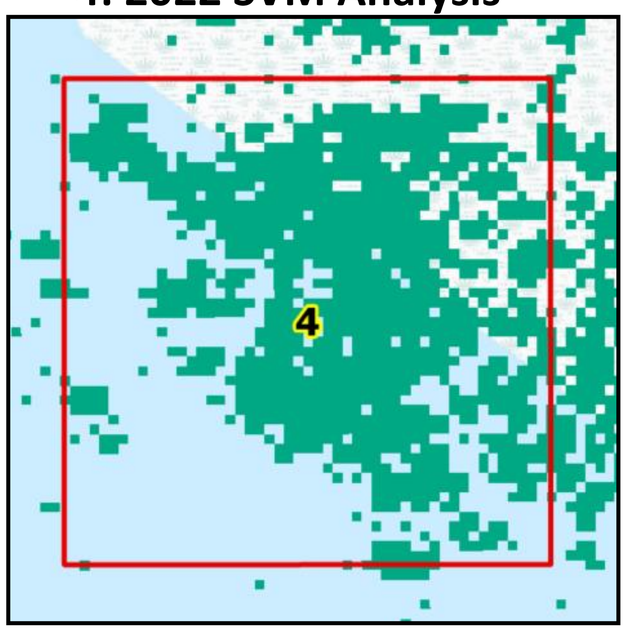
3: 2024 Visual NIR Image



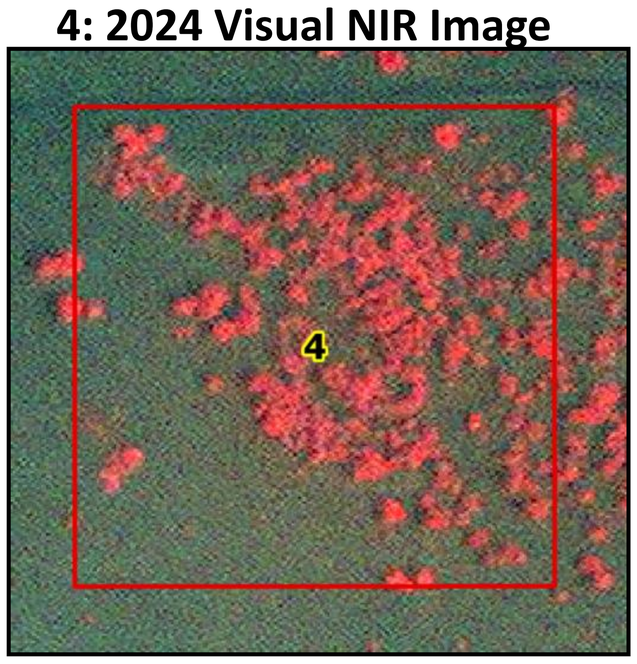
3: 2024 SVM Analysis



4: 2022 SVM Analysis



4: 2024 Visual NIR Image



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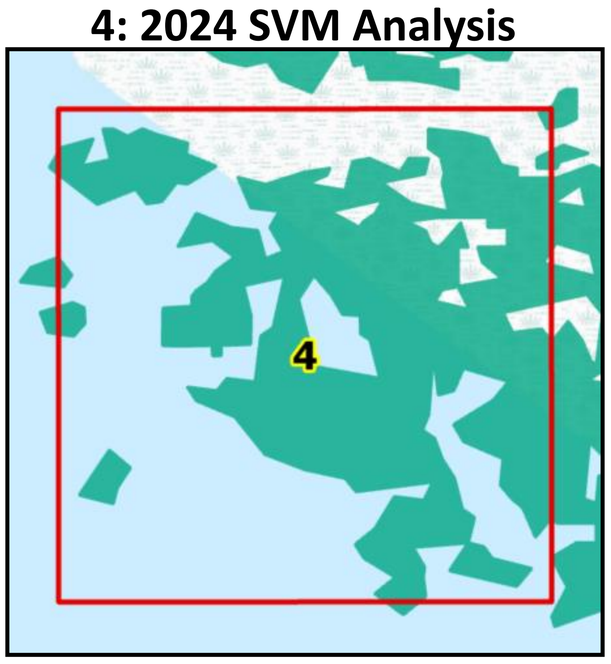
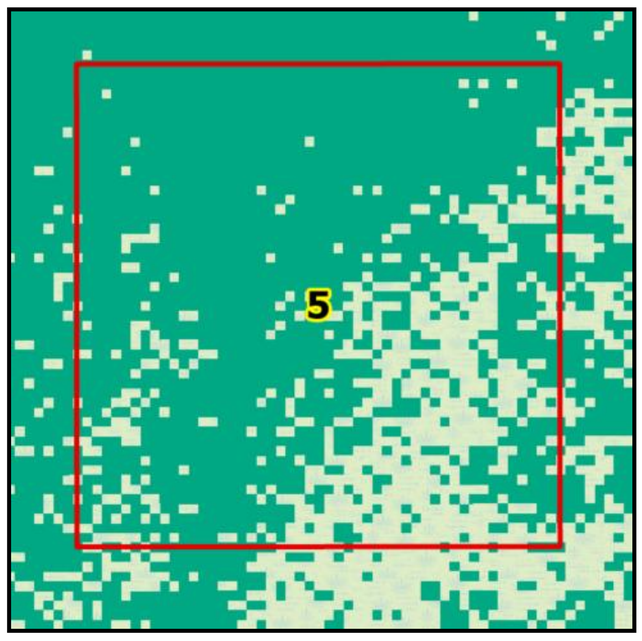
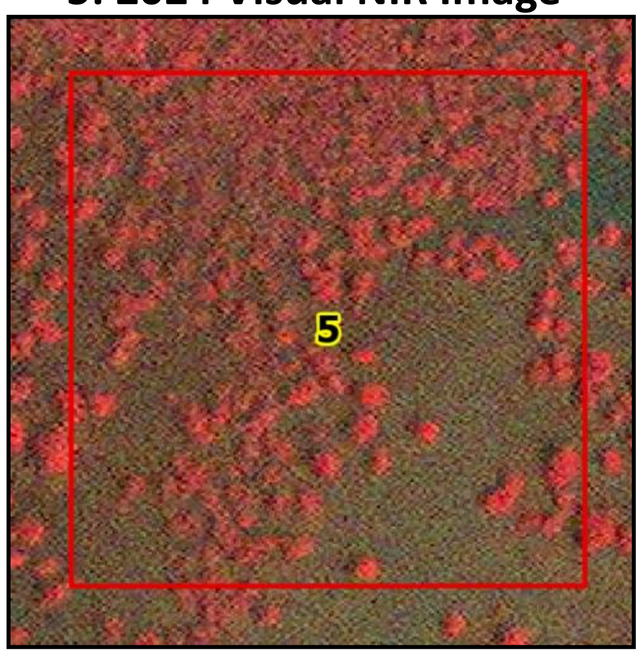


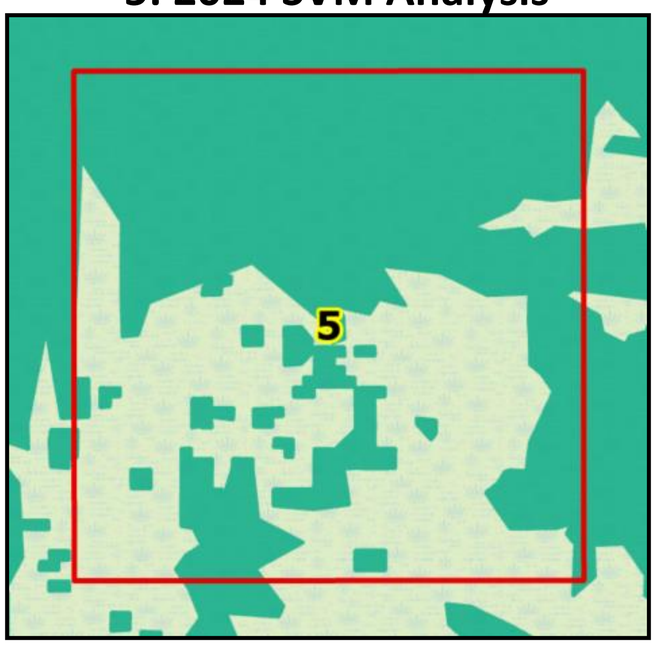
5: 2022 SVM Analysis



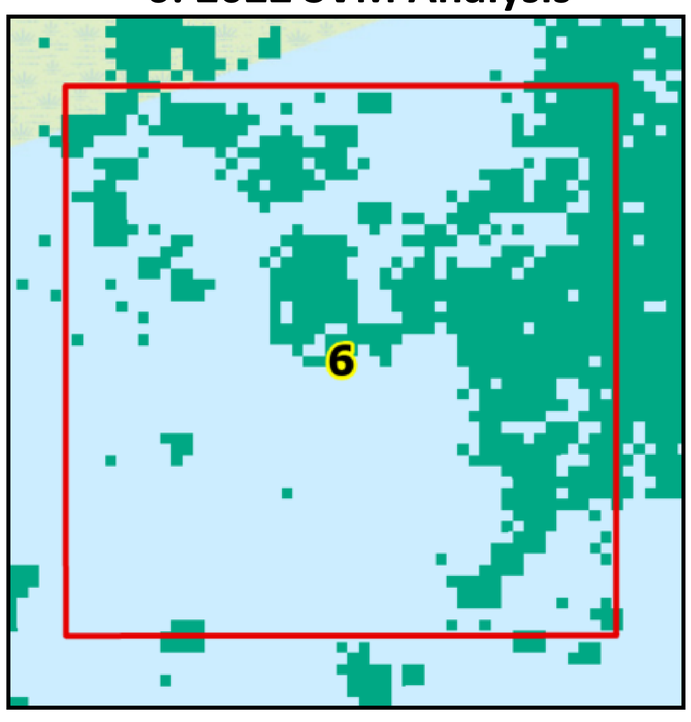
5: 2024 Visual NIR Image



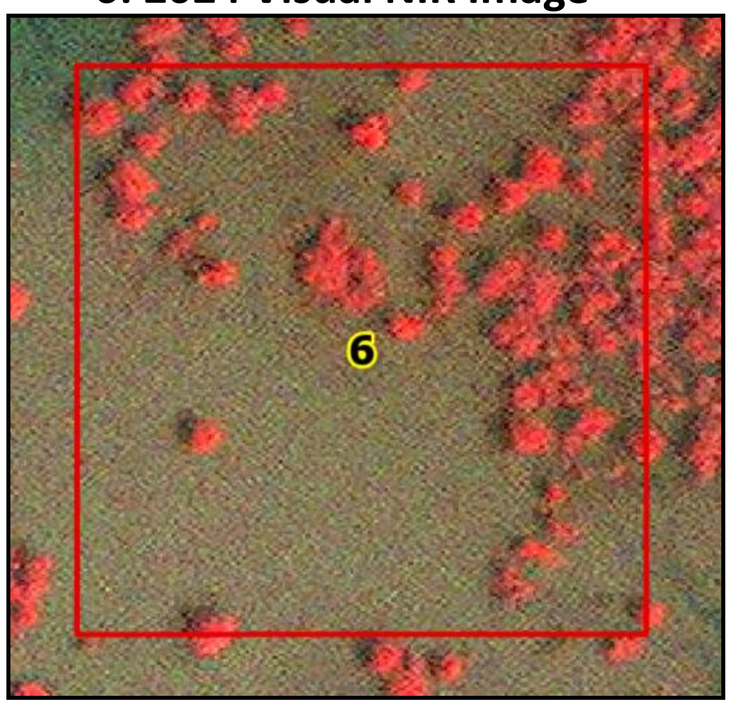
5: 2024 SVM Analysis



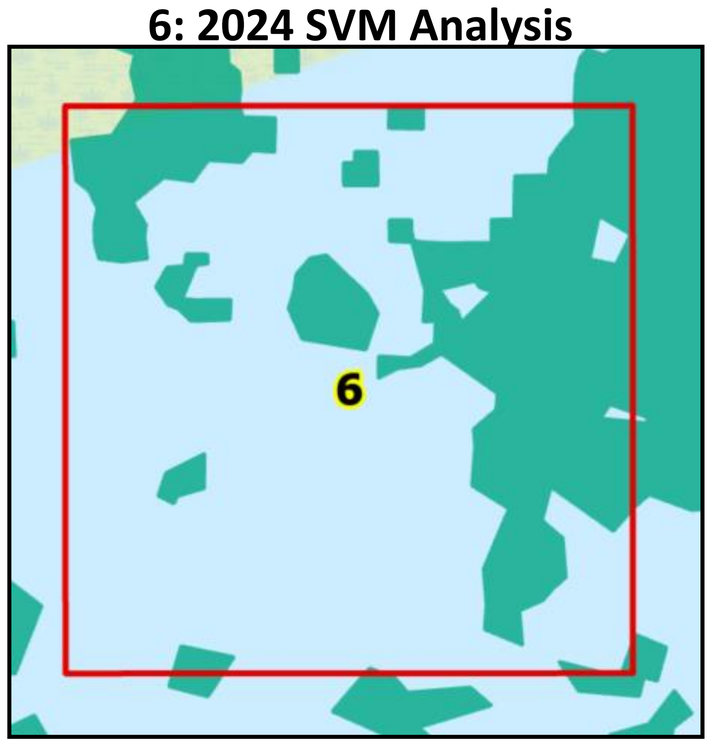
6: 2022 SVM Analysis



6: 2024 Visual NIR Image



6: 2024 SVM Analysis



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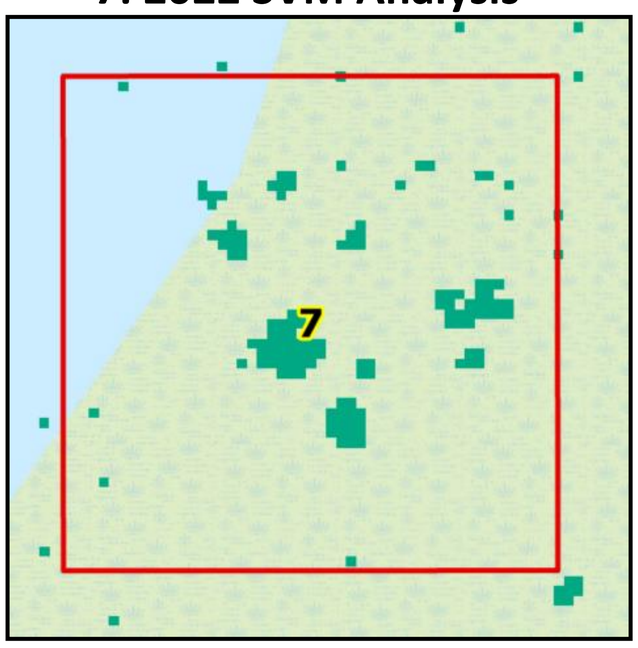
DATE: 23 May 2024

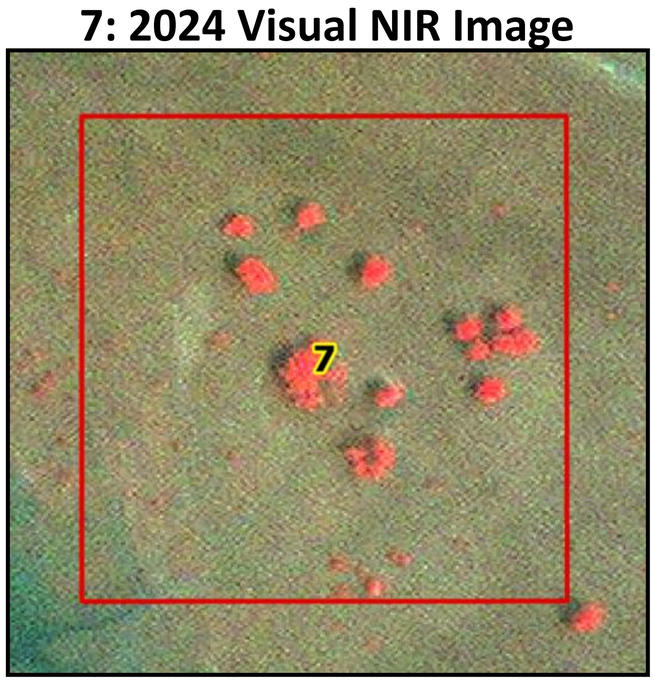
BODO REMEDIATION ANALYTICS-2024 LAND USE/LAND COVER STUDIES

Comparing Classification Results for 2022 SVM Analysis and 2024 SVM Analysis

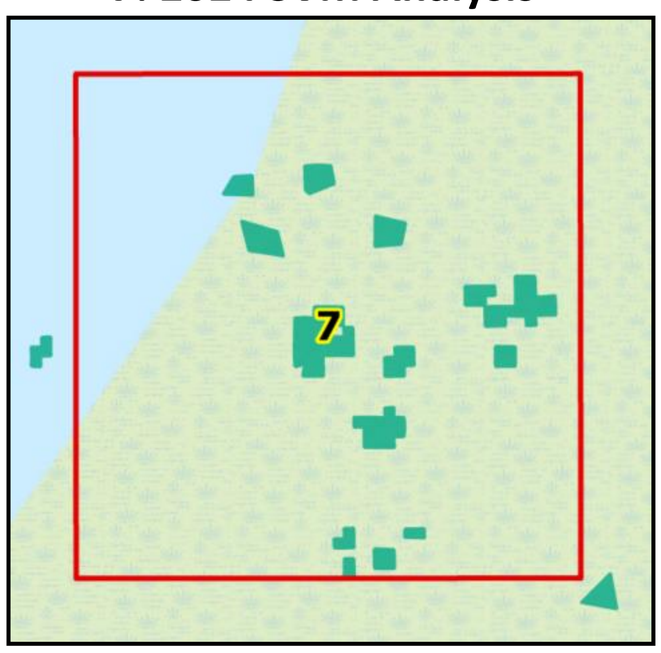
7: 2022 SVM Analysis



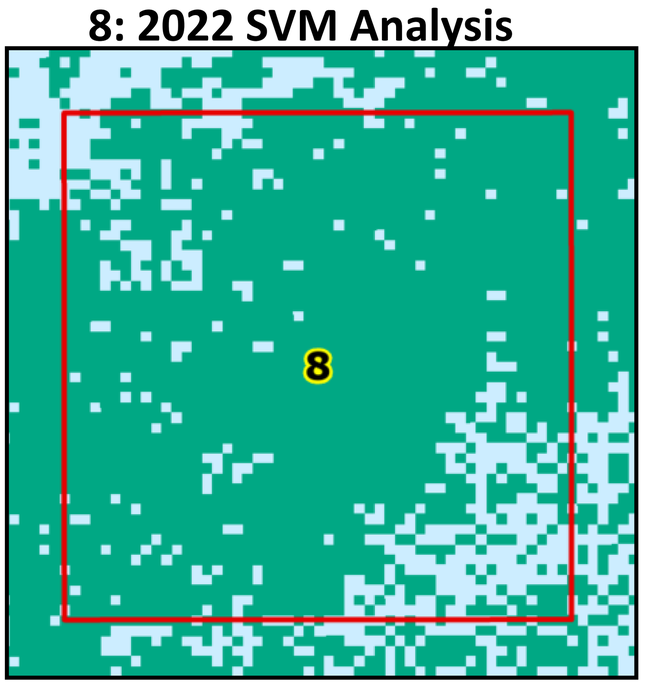
7: 2024 Visual NIR Image



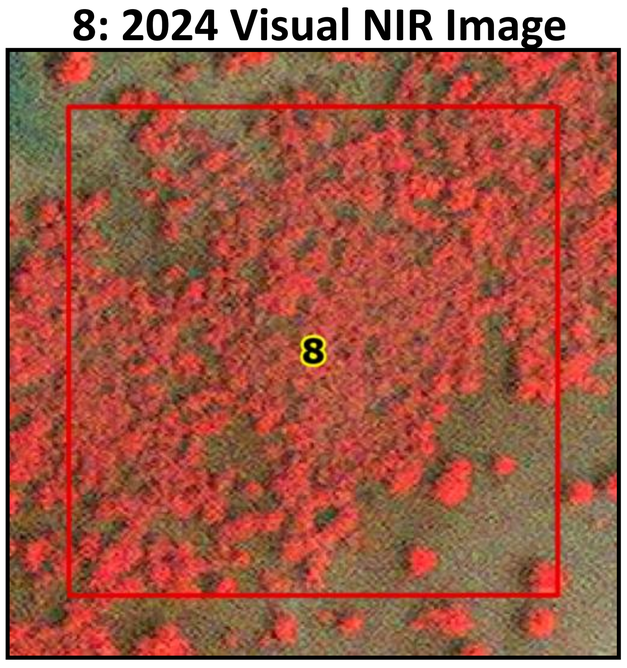
7: 2024 SVM Analysis



8: 2022 SVM Analysis



8: 2024 Visual NIR Image



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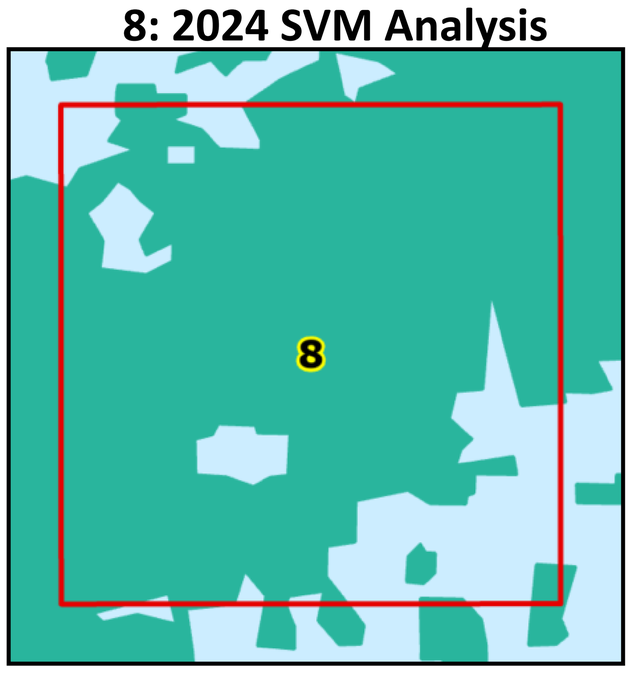
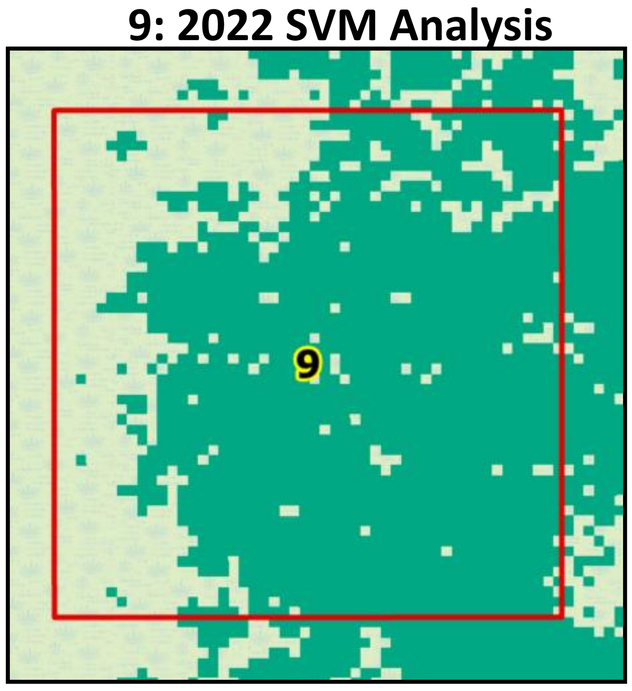


9: 2022 SVM Analysis



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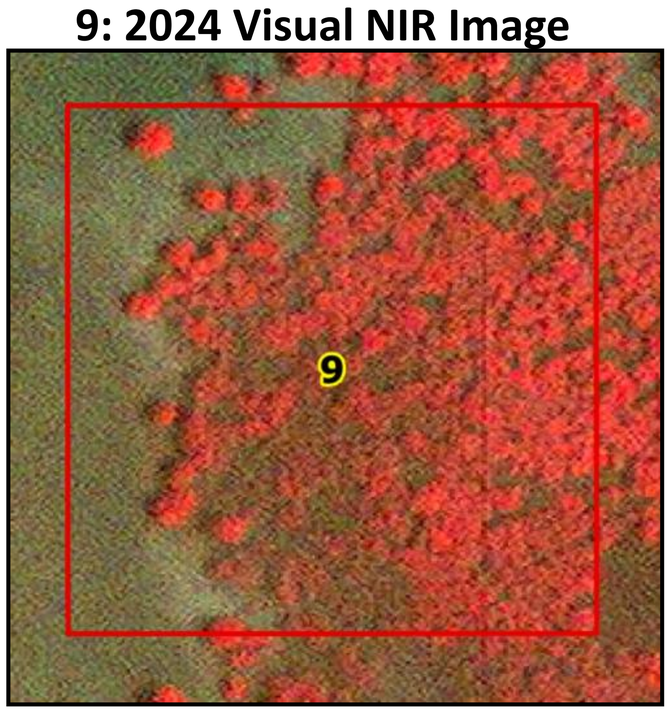
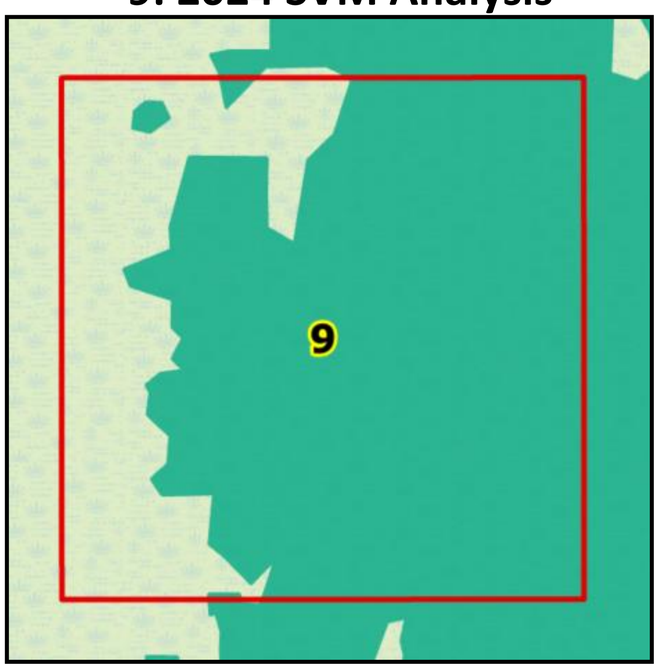
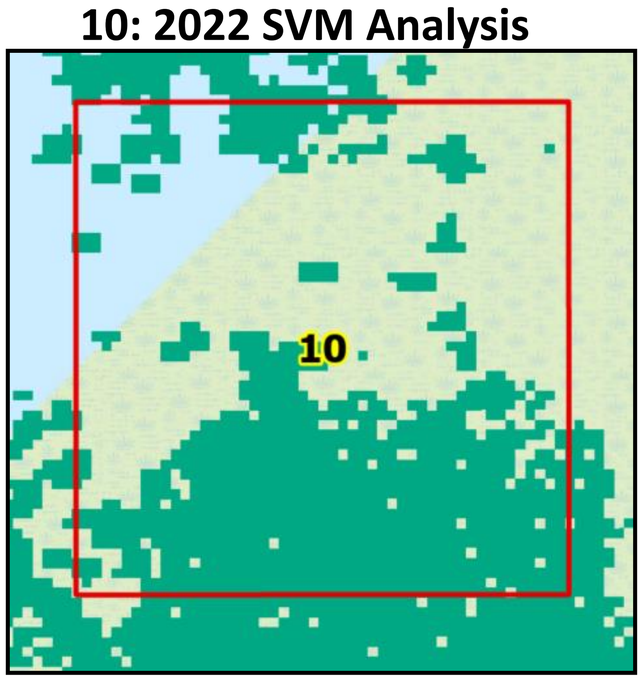


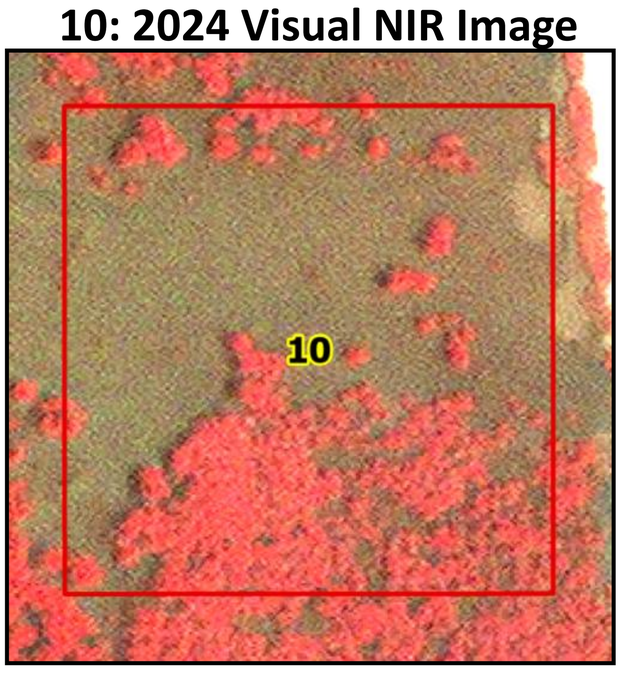
9: 2024 SVM Analysis



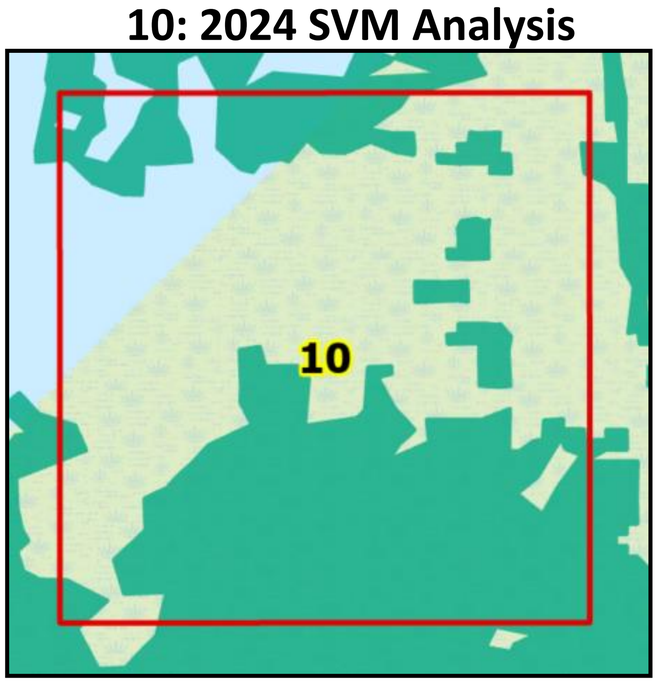
10: 2022 SVM Analysis



10: 2024 Visual NIR Image



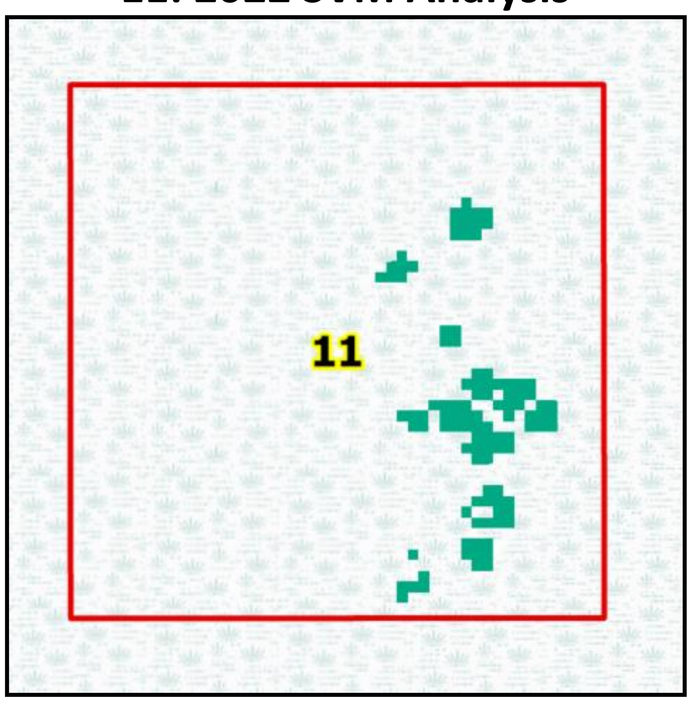
10: 2024 SVM Analysis



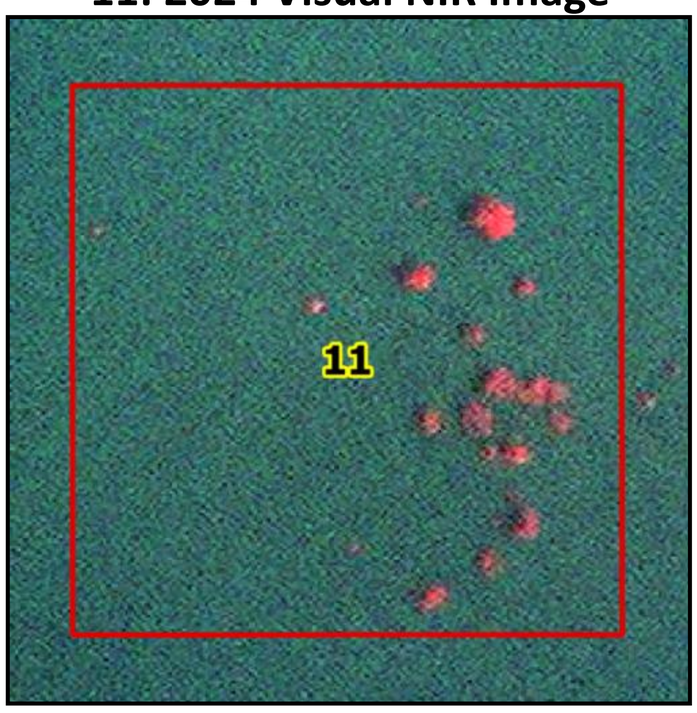
BODO REMEDIATION ANALYTICS-2024 LAND USE/LAND COVER STUDIES

Comparing Classification Results for 2022 SVM Analysis and 2024 SVM Analysis

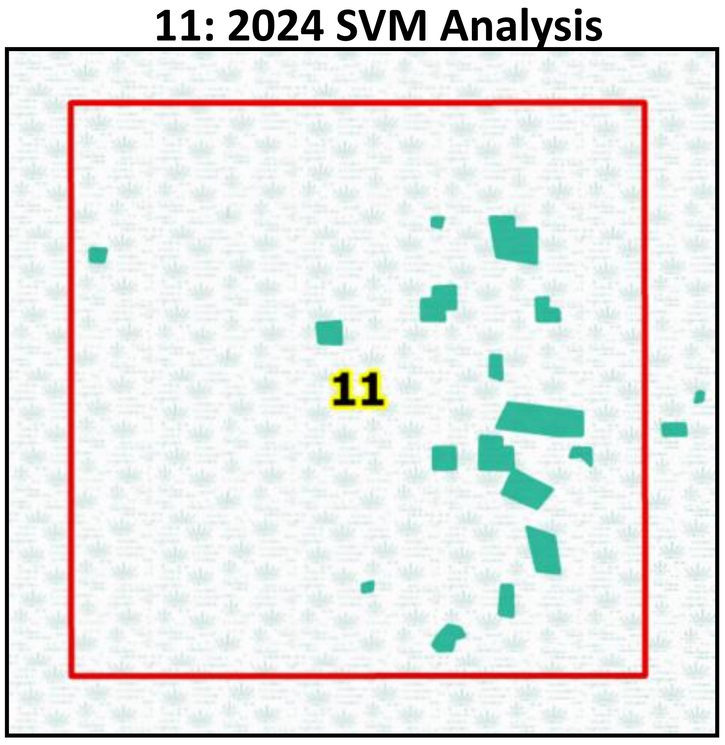
11: 2022 SVM Analysis



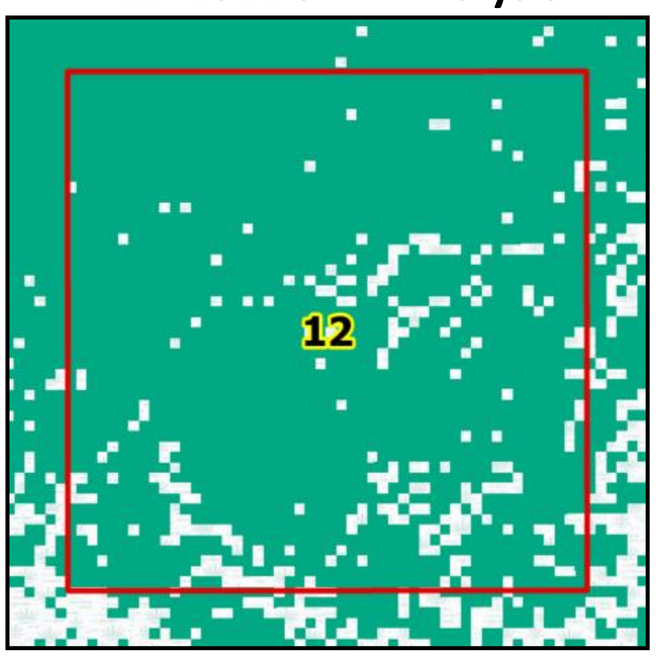
11: 2024 Visual NIR Image



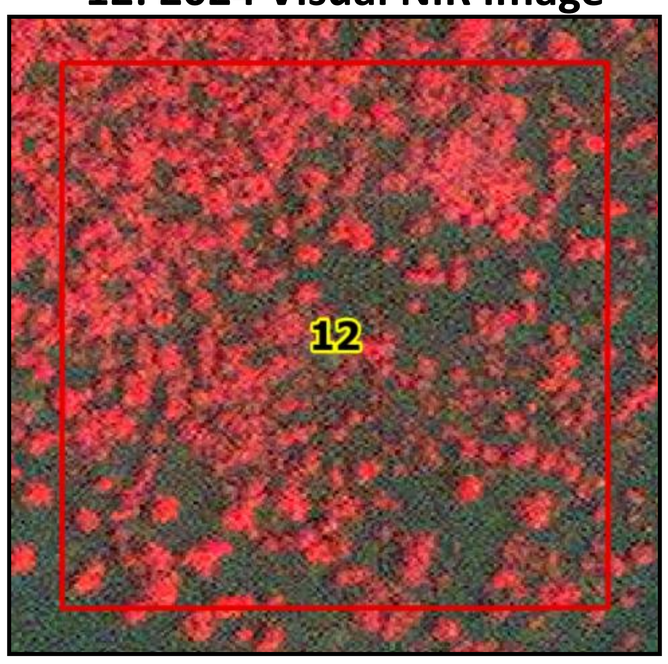
11: 2024 SVM Analysis



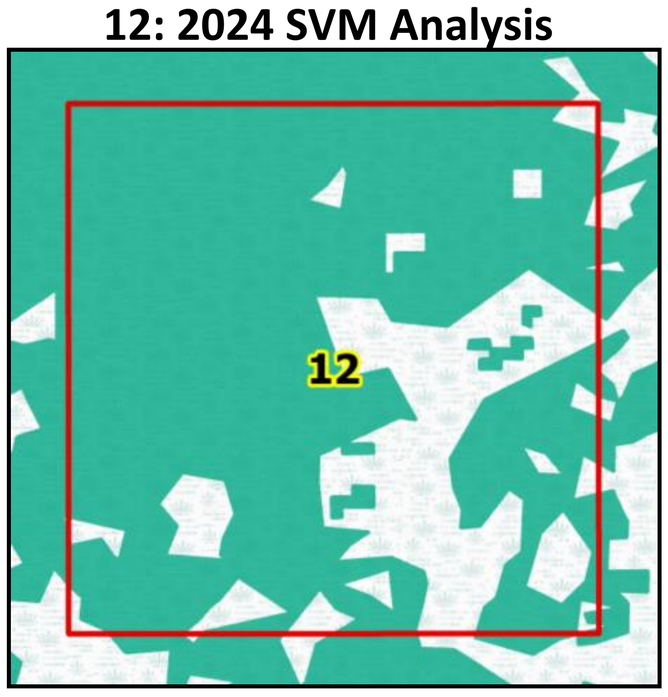
12: 2022 SVM Analysis



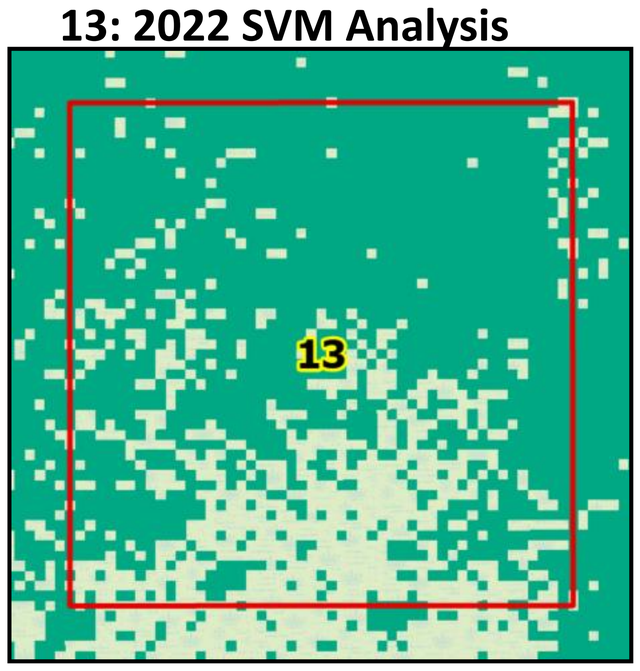
12: 2024 Visual NIR Image



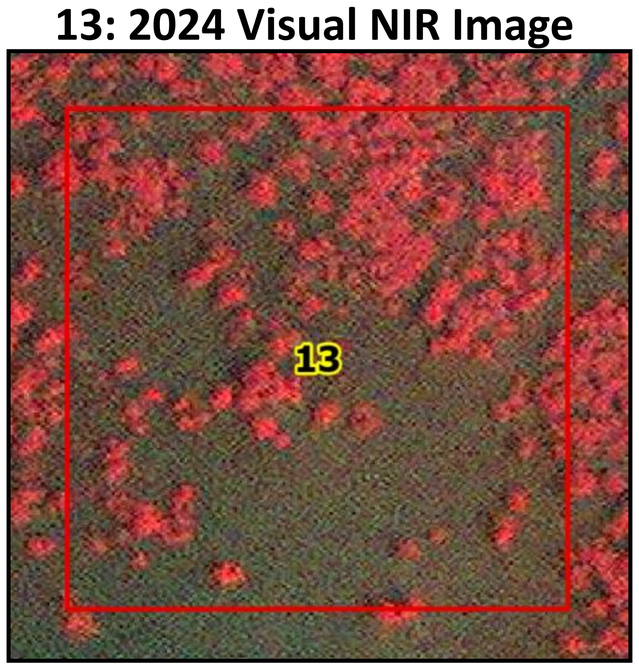
12: 2024 SVM Analysis



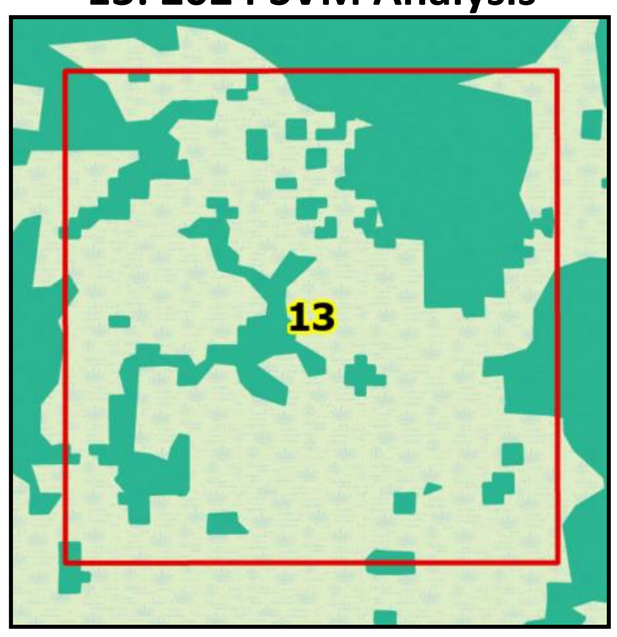
13: 2022 SVM Analysis



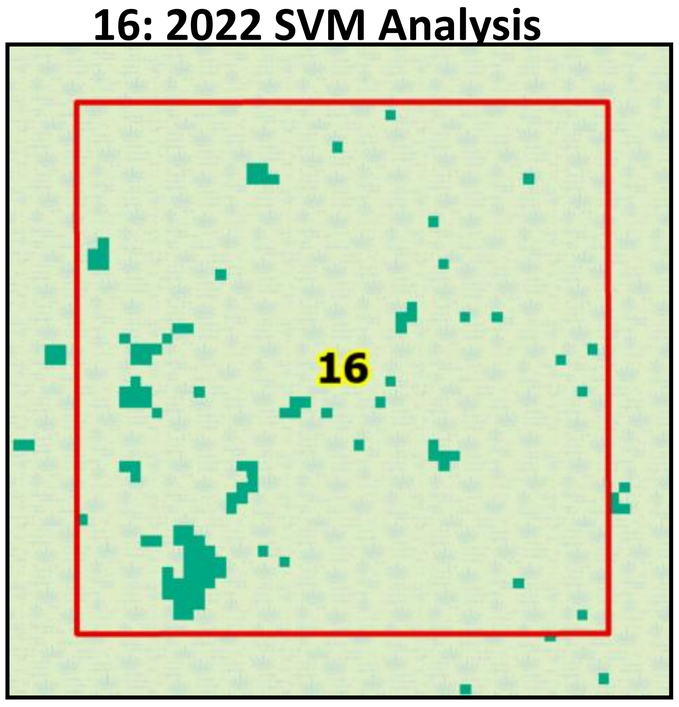
13: 2024 Visual NIR Image



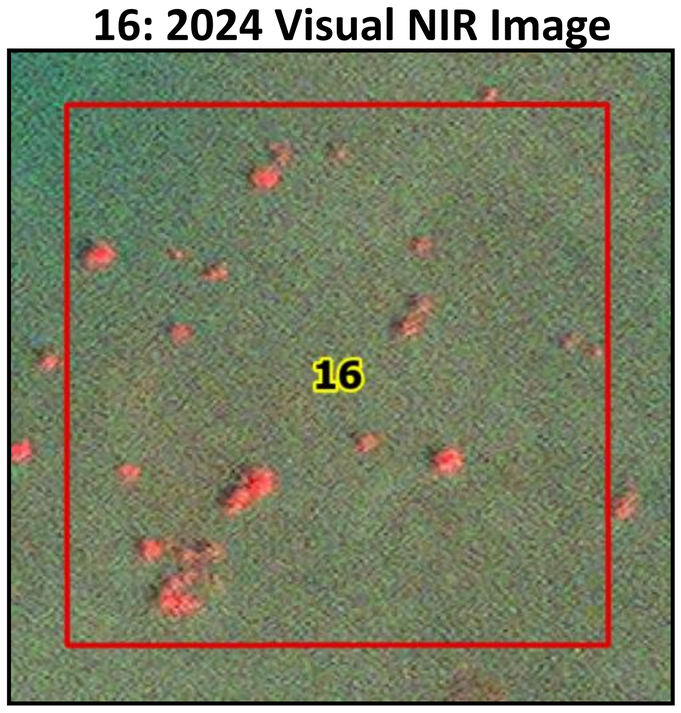
13: 2024 SVM Analysis



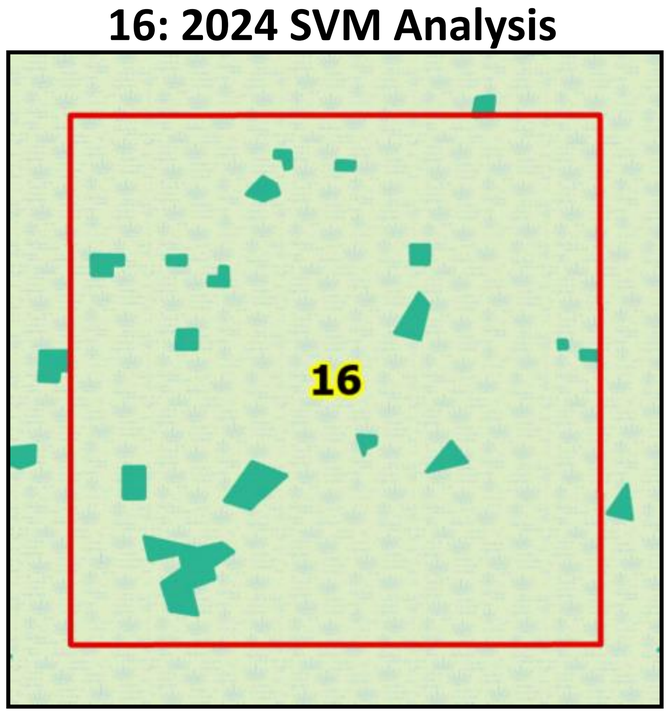
16: 2022 SVM Analysis



16: 2024 Visual NIR Image



16: 2024 SVM Analysis



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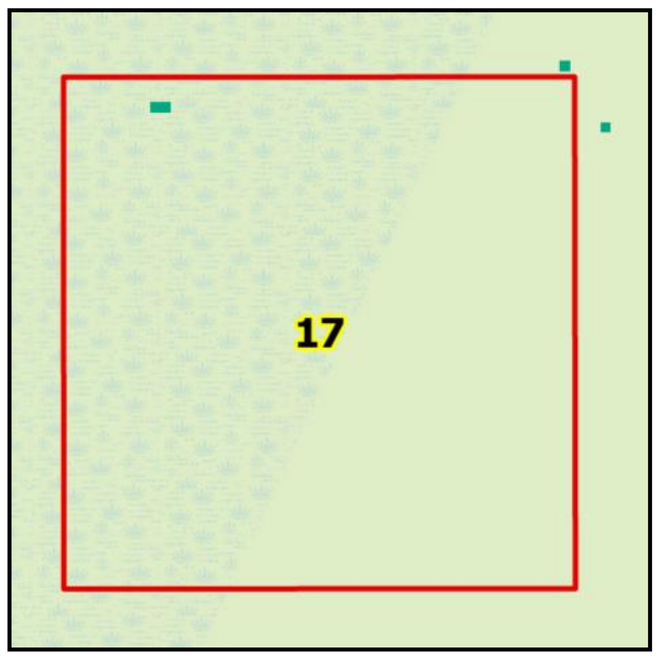
DATE: 23 May 2024

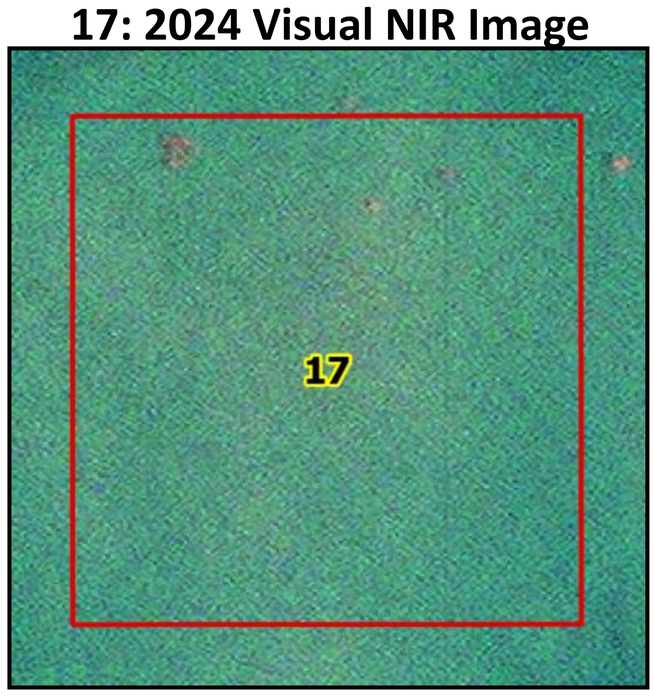
BODO REMEDIATION ANALYTICS-2024 LAND USE/LAND COVER STUDIES

Comparing Classification Results for 2022 SVM Analysis and 2024 SVM Analysis

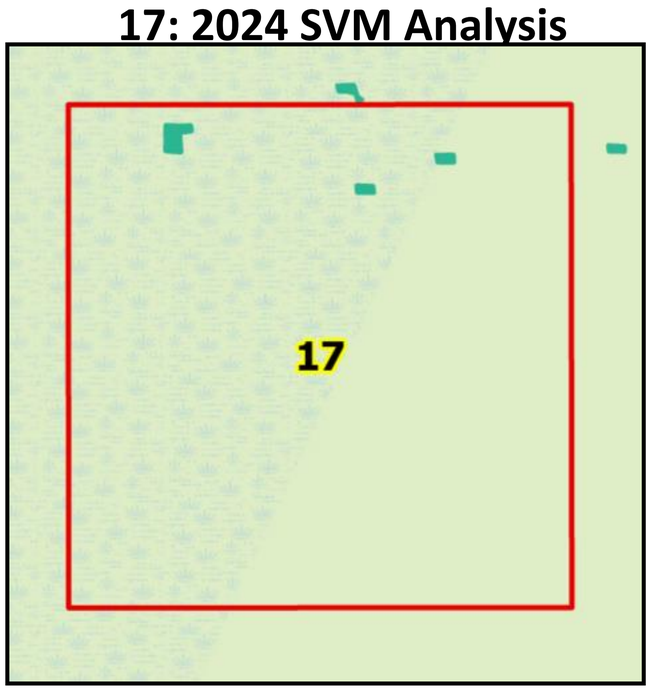
17: 2022 SVM Analysis



17: 2024 Visual NIR Image

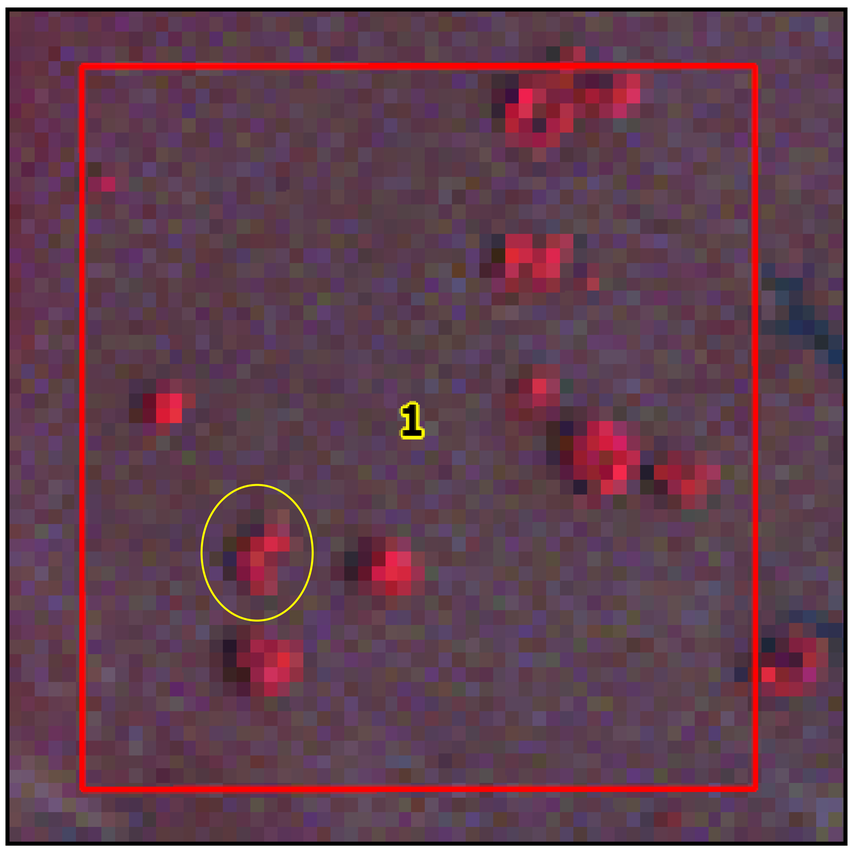


17: 2024 SVM Analysis

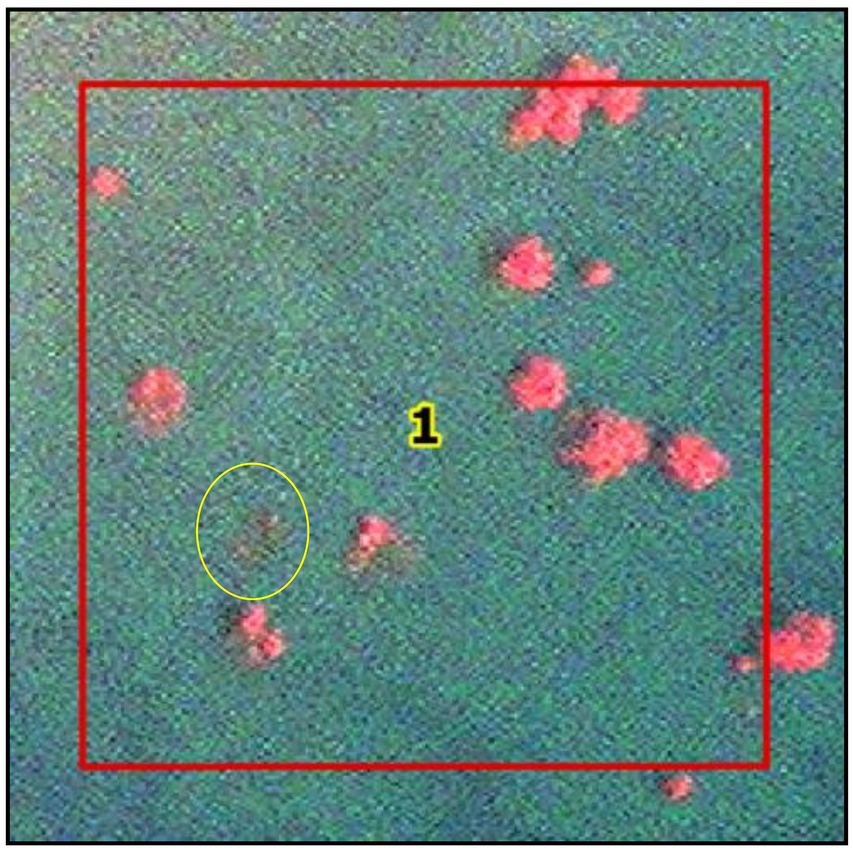


Change Detection Analysis Between 2022 Visual NIR and 2024 Visual NIR Images (Cells 1 – 13, 16-17)

1: 2022 Visual NIR Image

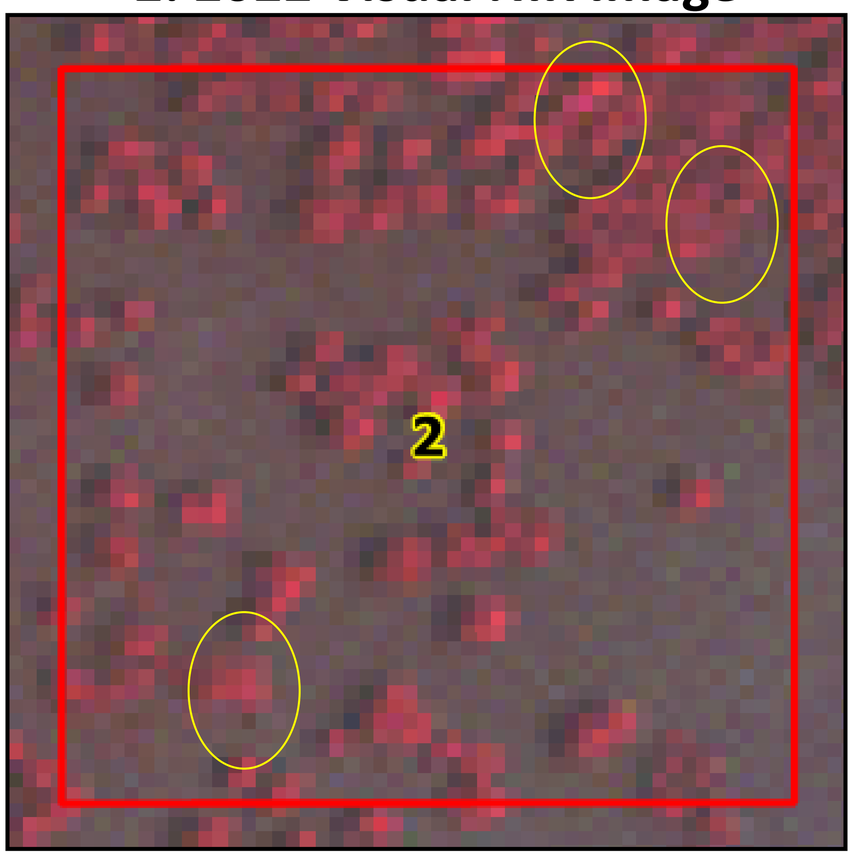


1: 2024 Visual NIR Image

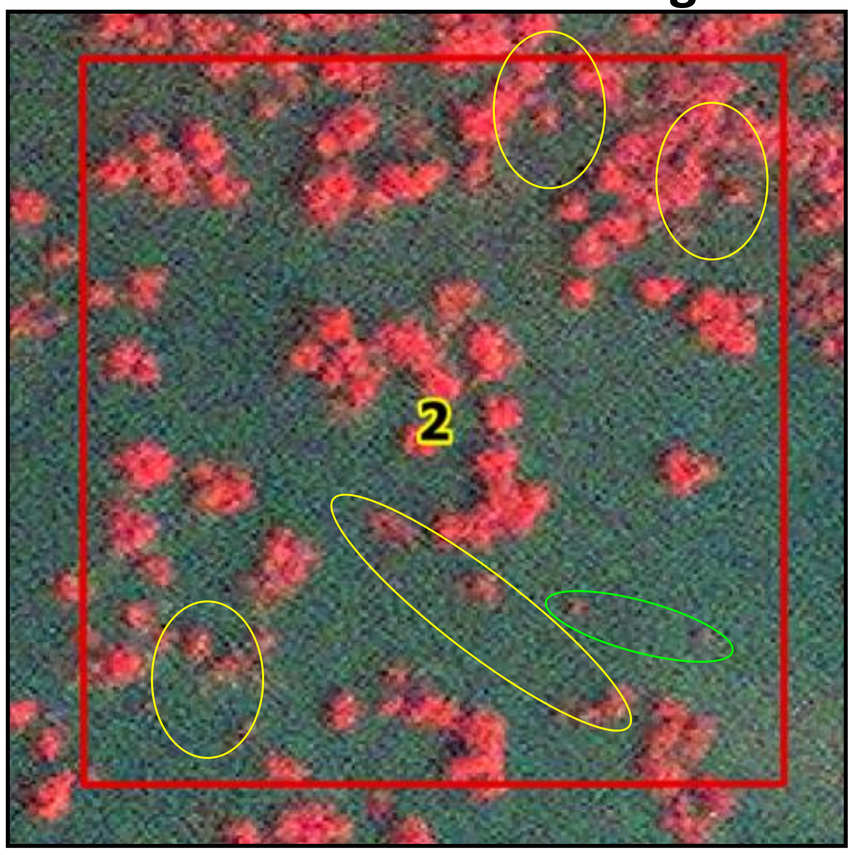


1 – There is not much difference observed between 2022 Visual NIR and the 2024 Visual NIR images. A noticeable reduction in vegetation foliage and size, circled in yellow, was observed in the 2024 image. Aside the highlighted vegetation loss, no obvious change noticed between 2022 – 2024 period.

2: 2022 Visual NIR Image



2: 2024 Visual NIR Image



2 – There is not much difference observed between the 2022 Visual NIR and the 2024 Visual NIR images. A noticeable reduction in vegetation foliage and size, circled in yellow, was observed in the 2024 image. This is most likely due to human or natural interference. However, there are some visible growth of vegetation highlighted within the green oval shape.

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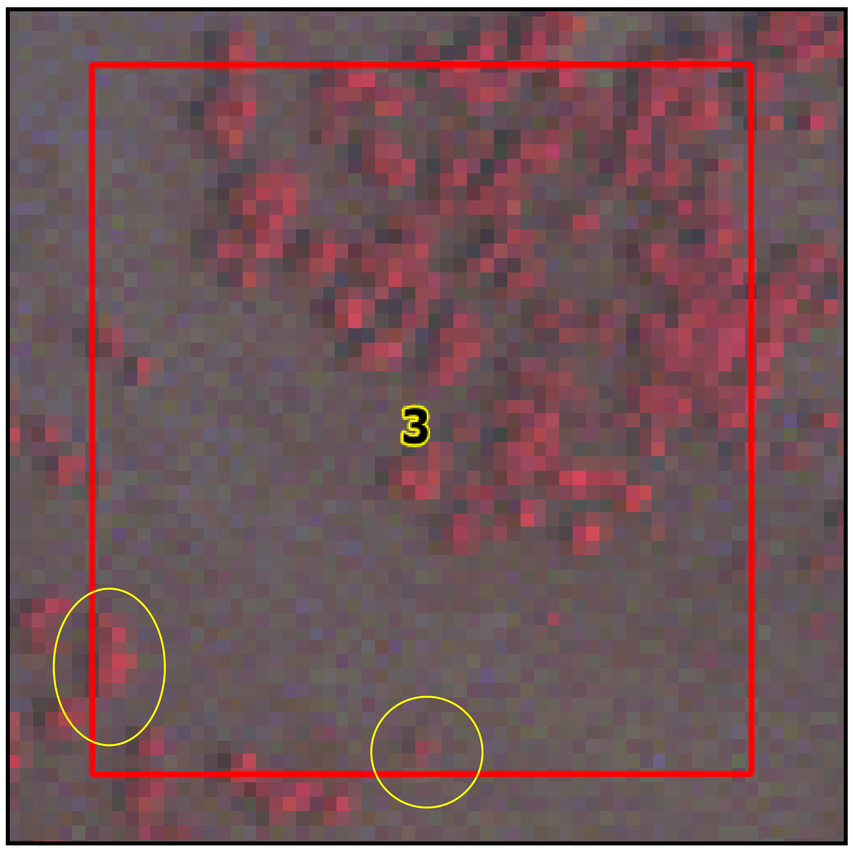
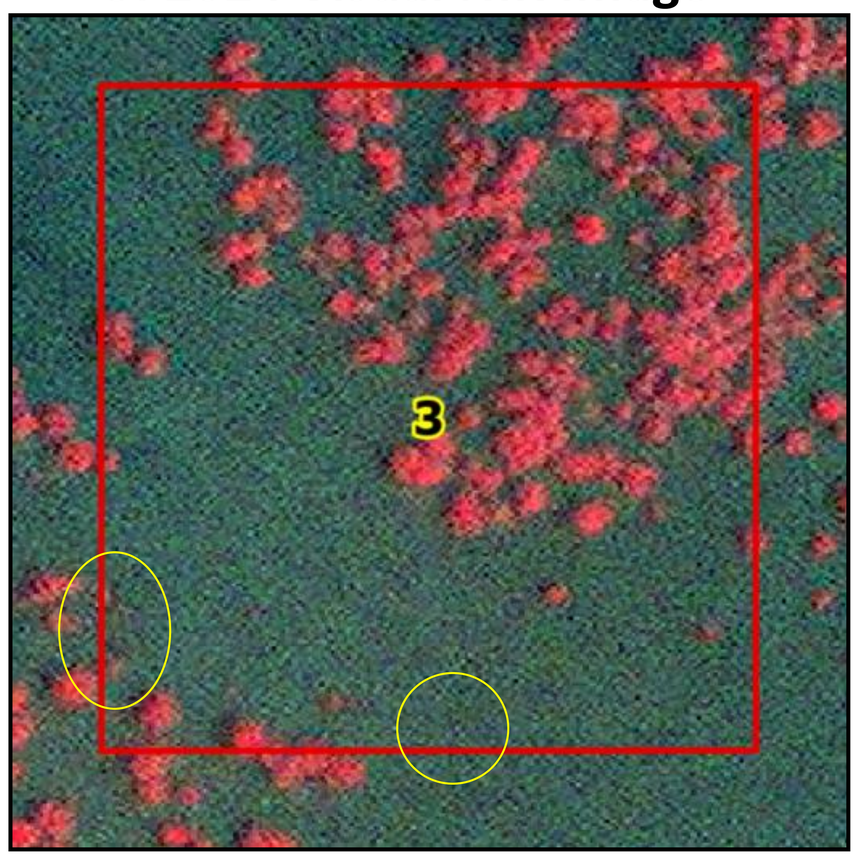
DATE: 23 May 2024

BODO REMEDIATION ANALYTICS - 2024 LAND USE/LAND COVER STUDIES

3: 2022 Visual NIR Image

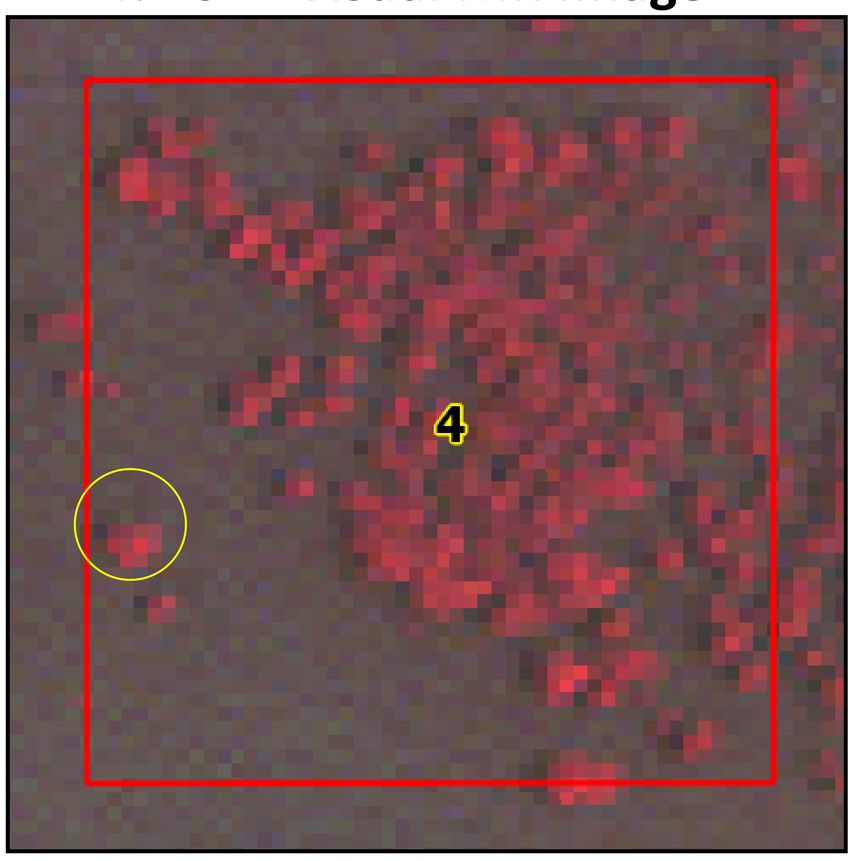


3: 2024 Visual NIR Image

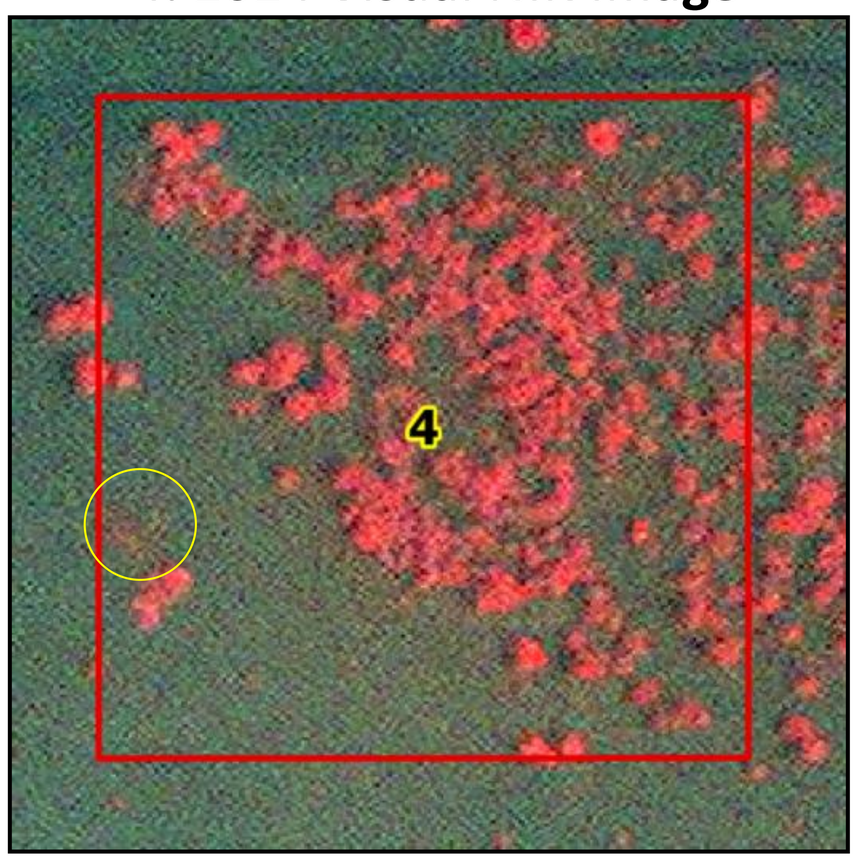


3 – The Pattern and distribution of vegetation appear similar between the 2022 Visual NIR and the 2024 Visual NIR images. A noticeable reduction in vegetation foliage and size, circled in yellow, was observed in the 2024 image. Aside the highlighted vegetation loss, no obvious change noticed between 2022 – 2024 period.

4: 2022 Visual NIR Image

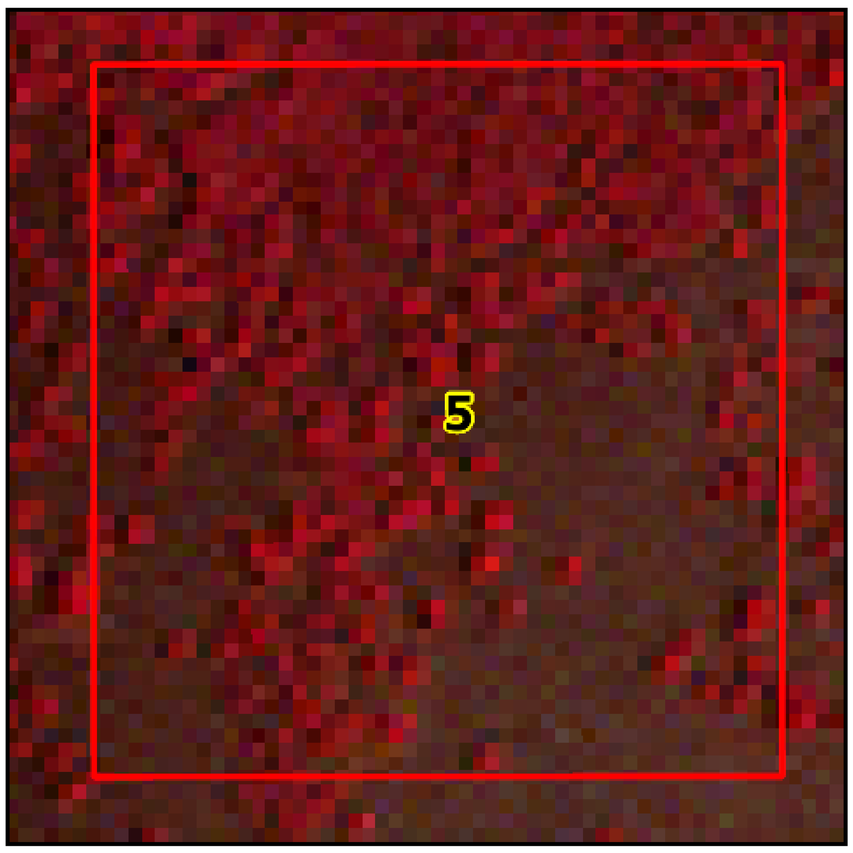


4: 2024 Visual NIR Image

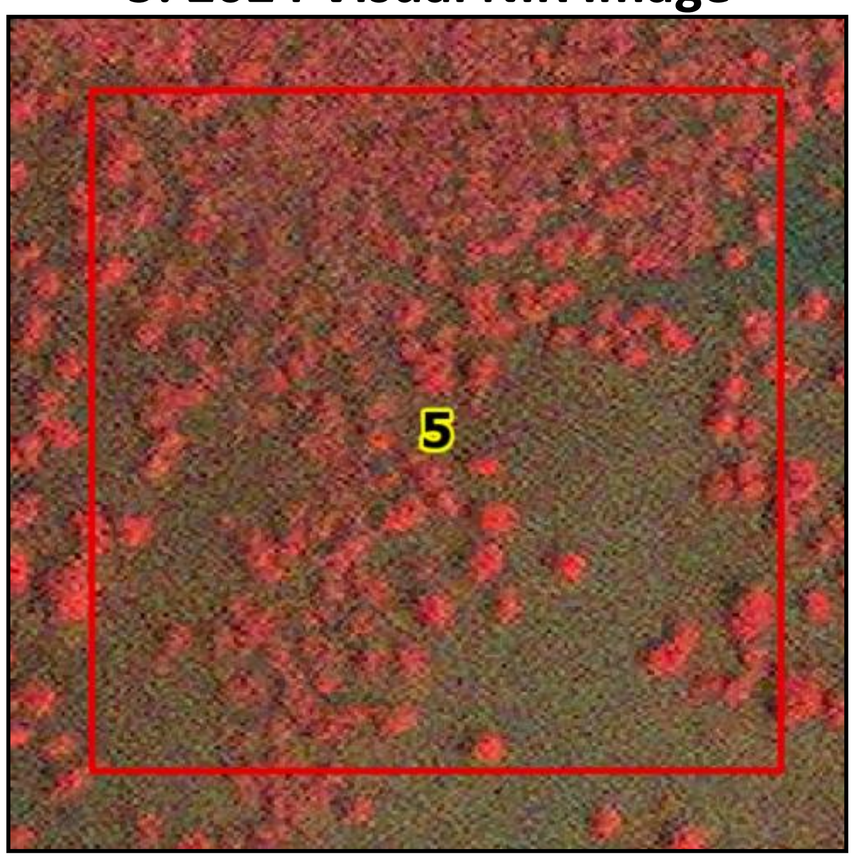


4 – Cell 4 has similar vegetation pattern and distribution between 2022 and 2024 period. There are no significant growth observed within the 2year period. Loss of vegetation highlighted within the yellow circle.

5: 2022 Visual NIR Image



5: 2024 Visual NIR Image



5 – The Pattern and distribution of vegetation appear similar between the 2022 Visual NIR and the 2024 Visual NIR images. There are no significant mangrove growth or difference observed.

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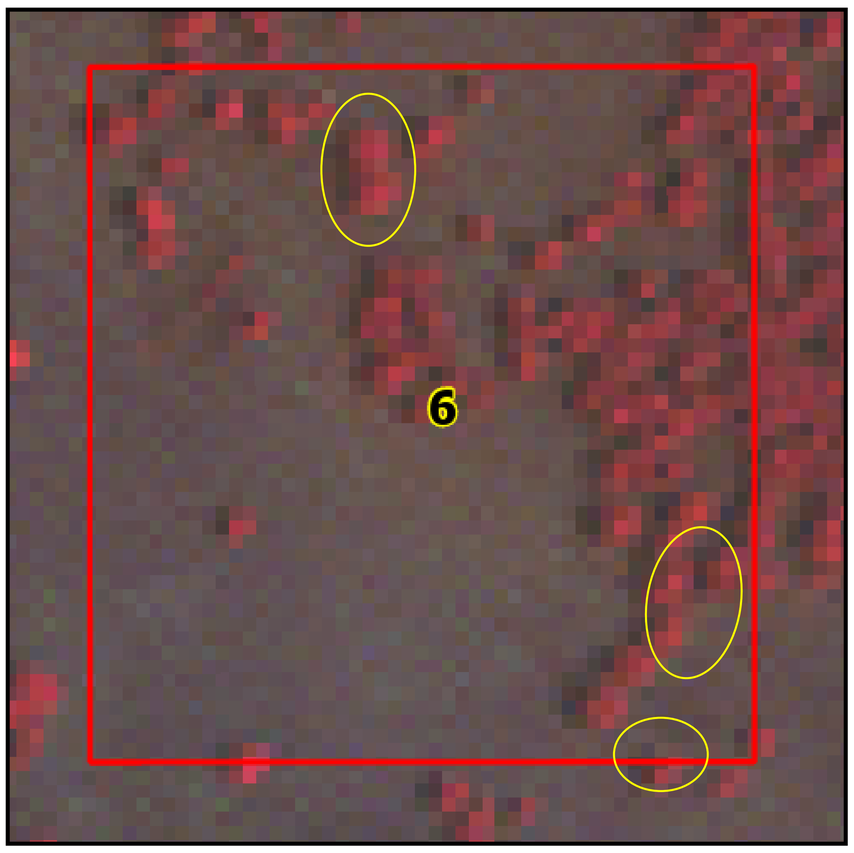
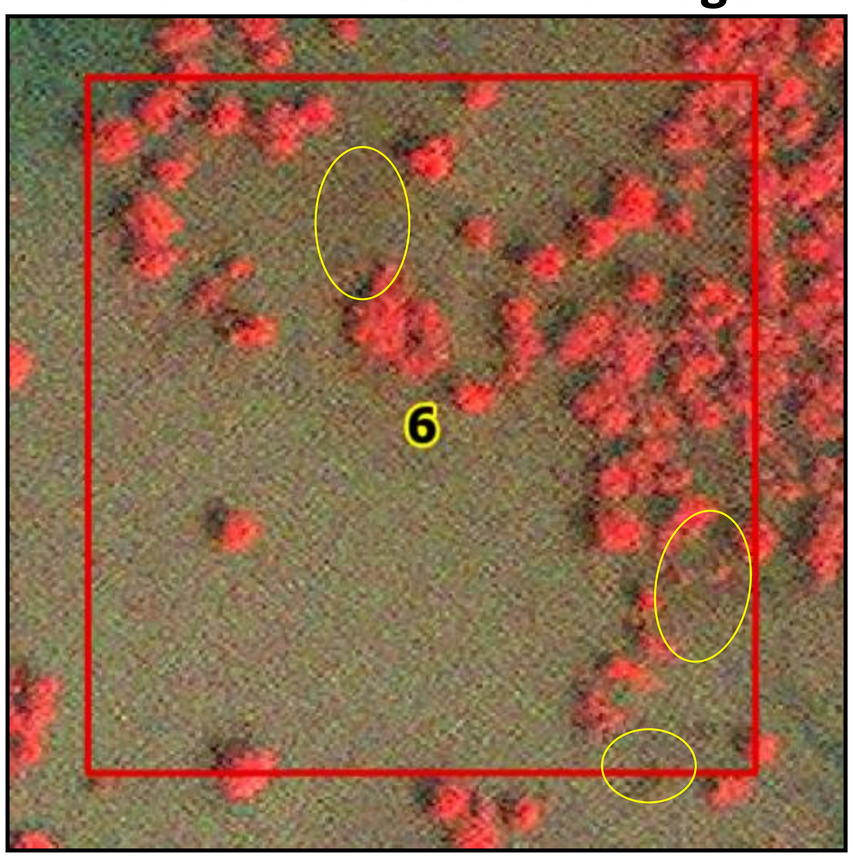
BODO REMEDIATION ANALYTICS-2024 LAND USE/LAND COVER STUDIES

6: 2022 Visual NIR Image

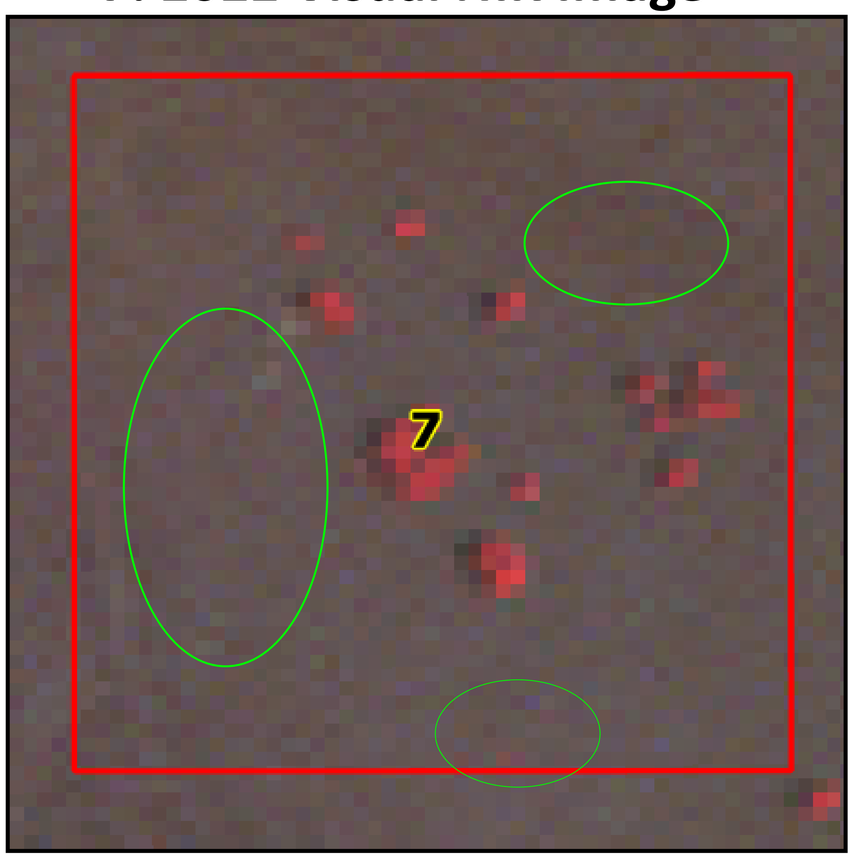


6: 2024 Visual NIR Image

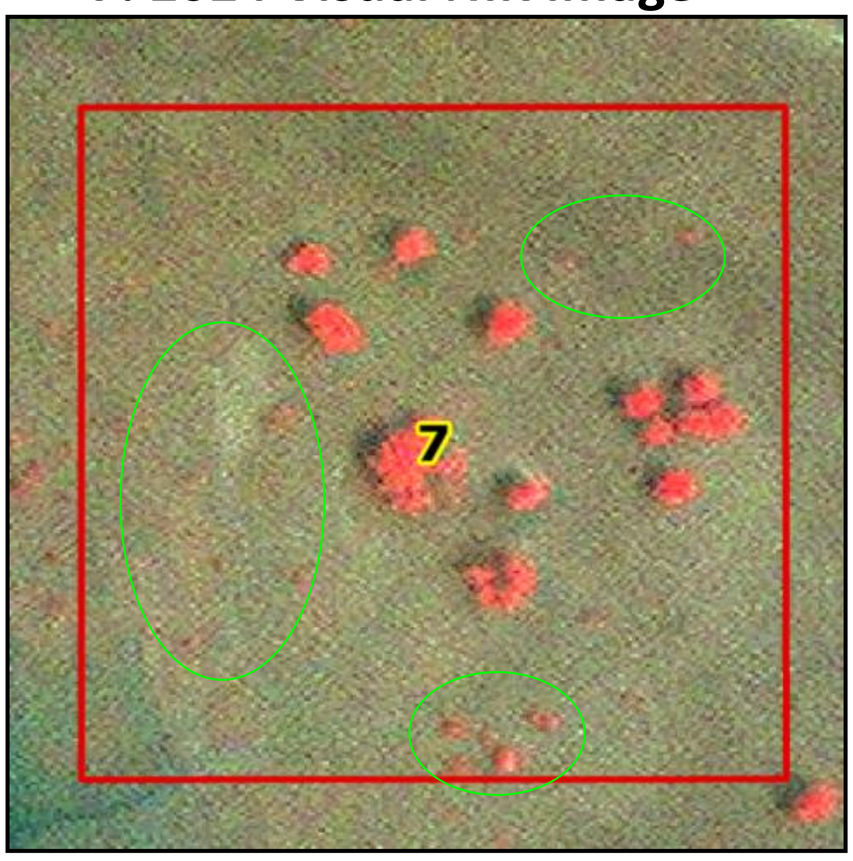


6 – The distribution and pattern of vegetation appear similar. A noticeable reduction in vegetation foliage and size, circled in yellow, was observed in the 2024 image.

7: 2022 Visual NIR Image

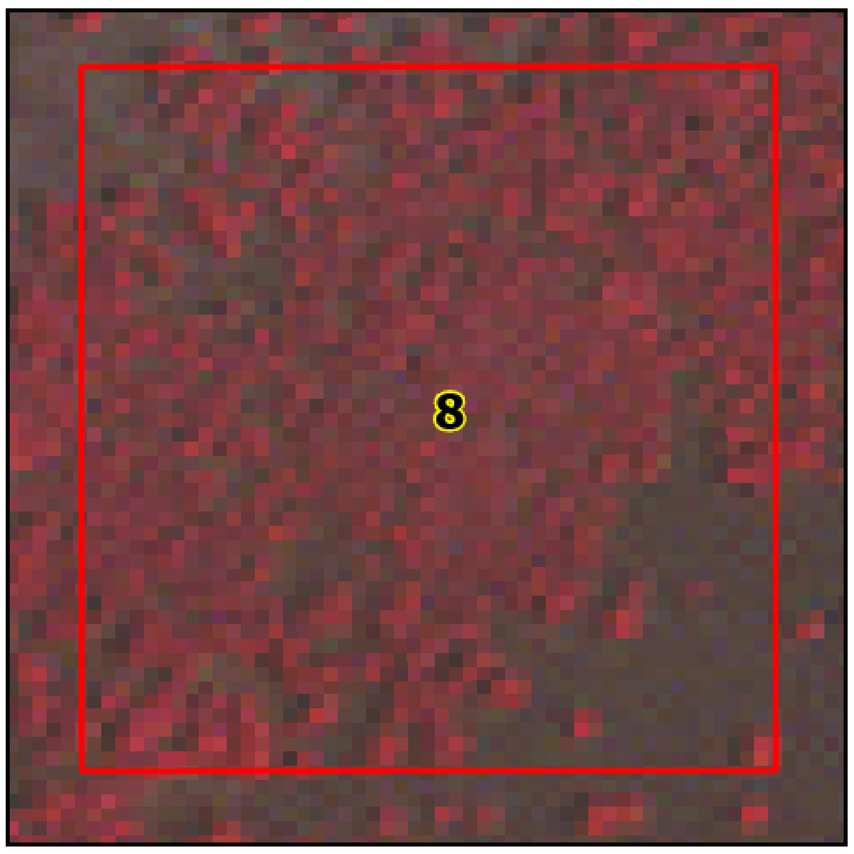


7: 2024 Visual NIR Image

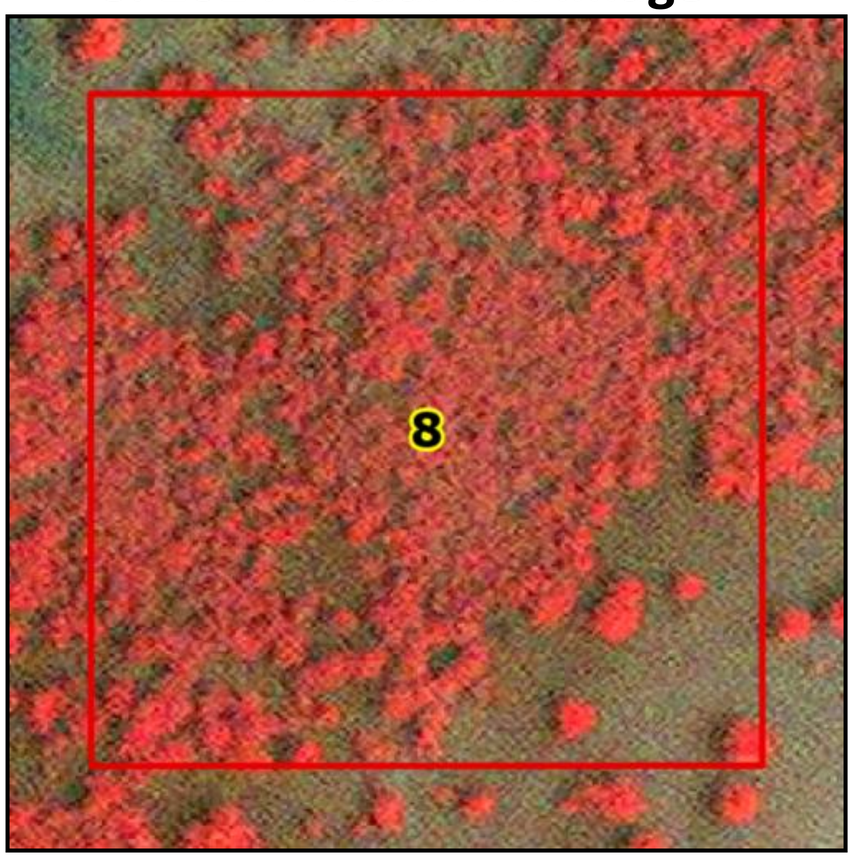


7 – There are indications of vegetation sprouting in the areas highlighted within the green oval shapes. Distribution of the obvious grown vegetation appear similar.

8: 2022 Visual NIR Image



8: 2024 Visual NIR Image



8 - The distribution and pattern of vegetation appear similar between both year of study. There are no significant growth or difference observed.

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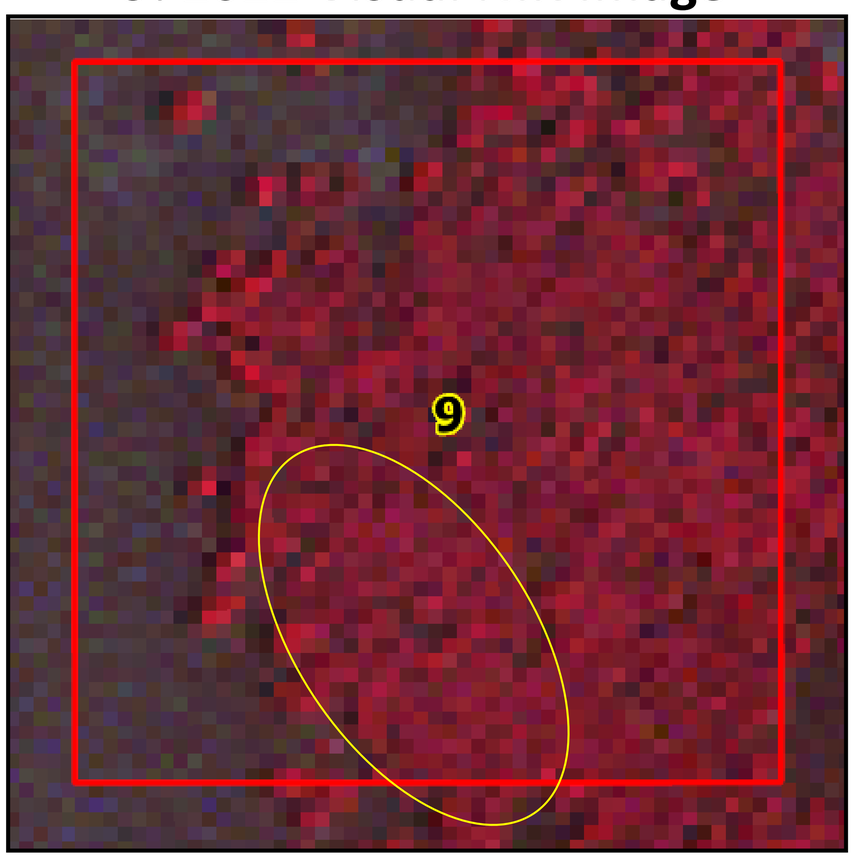
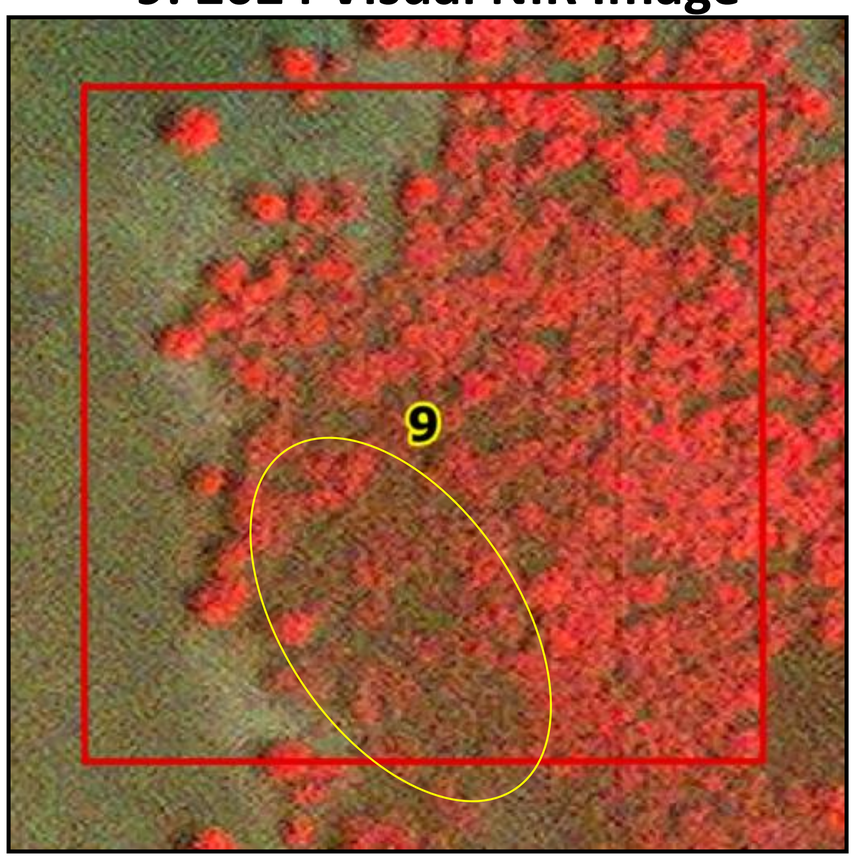
BODO REMEDIATION ANALYTICS-2024 LAND USE/LAND COVER STUDIES

9: 2022 Visual NIR Image

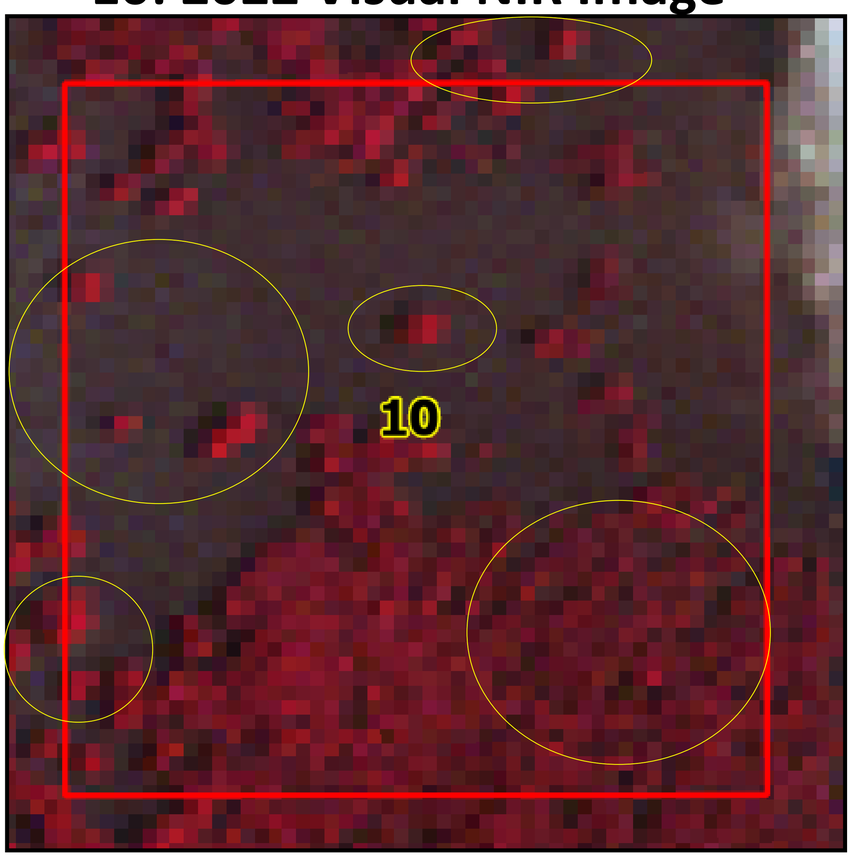


9: 2024 Visual NIR Image

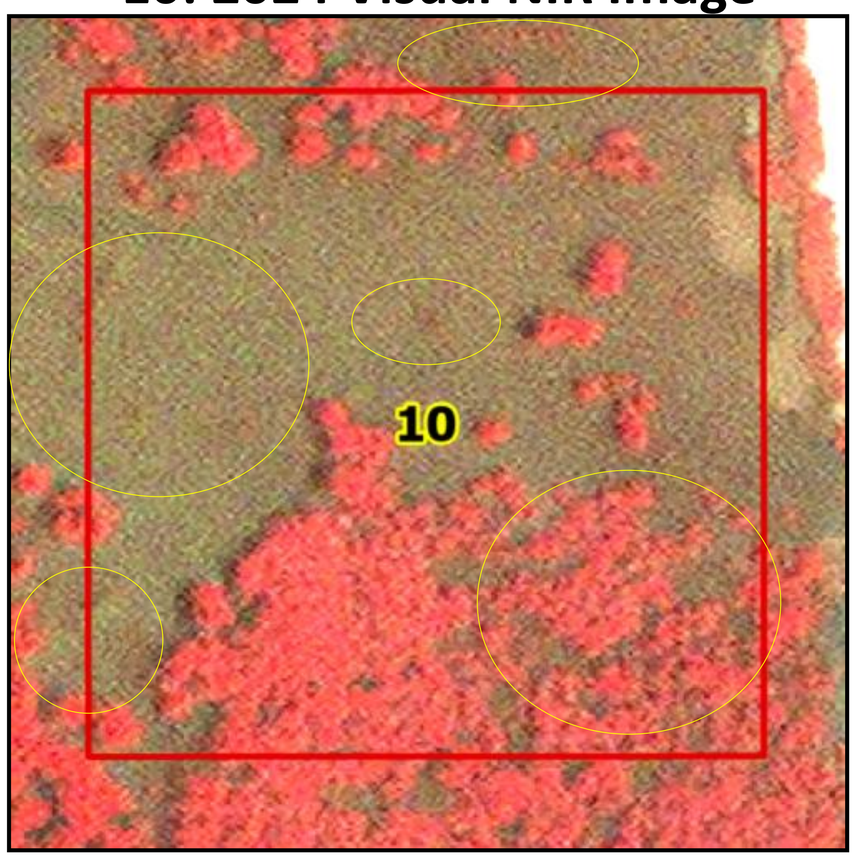


9 - The Pattern and distribution of vegetation appear similar between the 2022 Visual NIR and the 2024 Visual NIR images. However, there is a noticeable defoliation of the vegetation, circled in yellow, observed in the 2024 image. Aside the highlighted vegetation loss, no obvious change noticed between 2022 – 2024 period.

10: 2022 Visual NIR Image

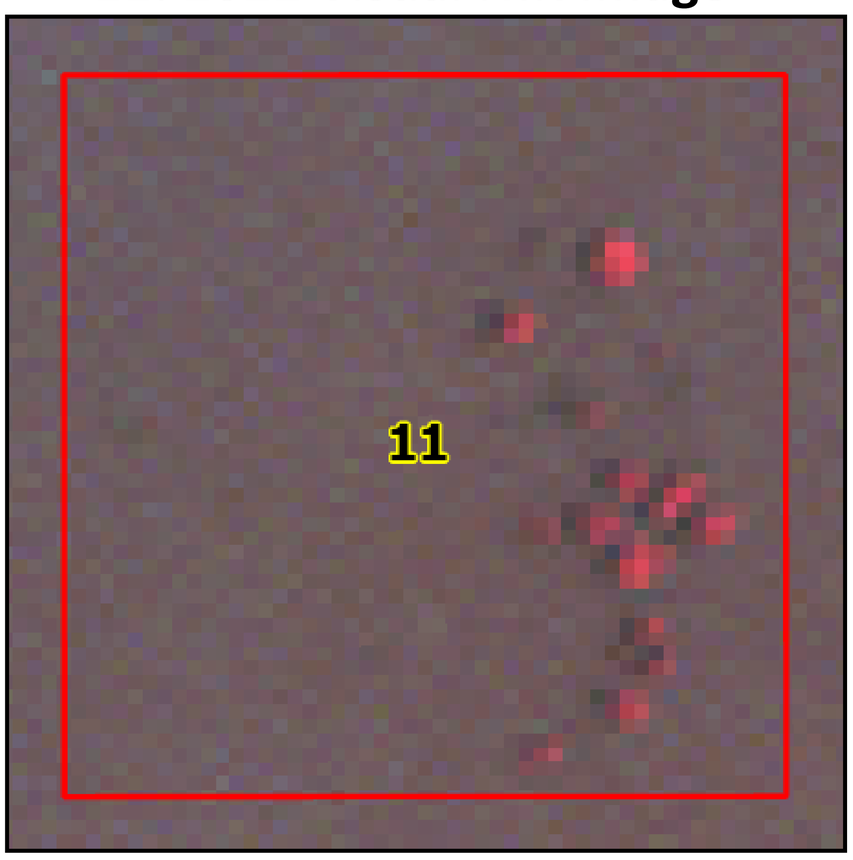


10: 2024 Visual NIR Image

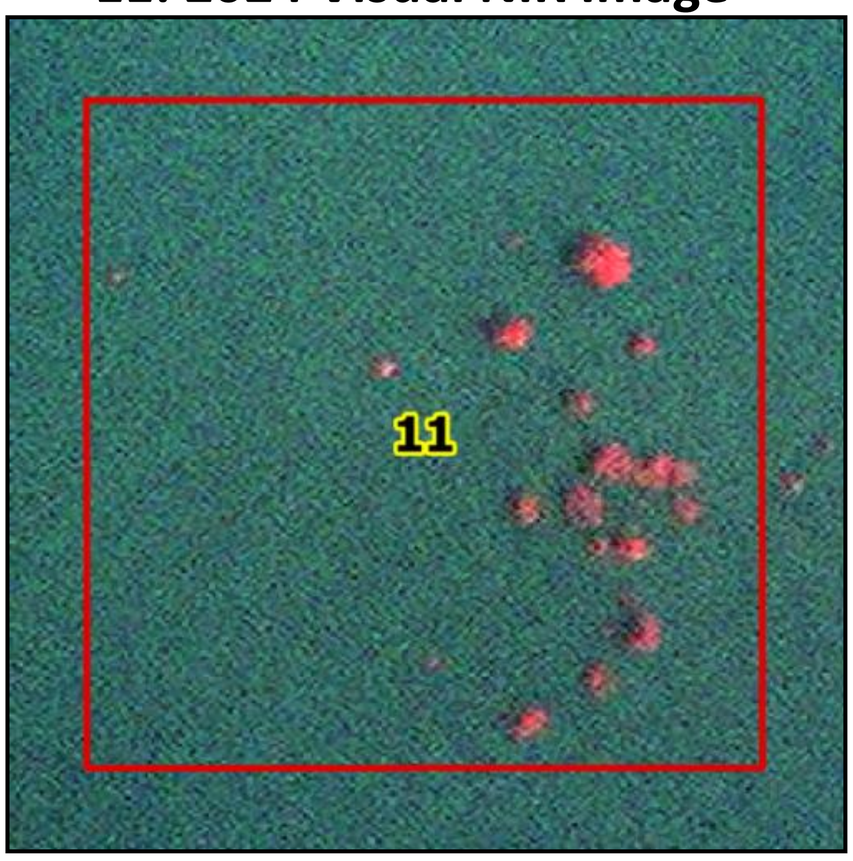


10 - There is little difference observed between the 2022 Visual NIR and the 2024 Visual NIR images. A noticeable disappearance and reduction in vegetation foliage and size, circled in yellow, is observed in the 2024 image. This is most likely due to human or natural interference.

11: 2022 Visual NIR Image



11: 2024 Visual NIR Image



11 - The Pattern and distribution of vegetation appear similar between the 2022 Visual NIR and the 2024 Visual NIR images. There are no significant mangrove growth or difference observed.

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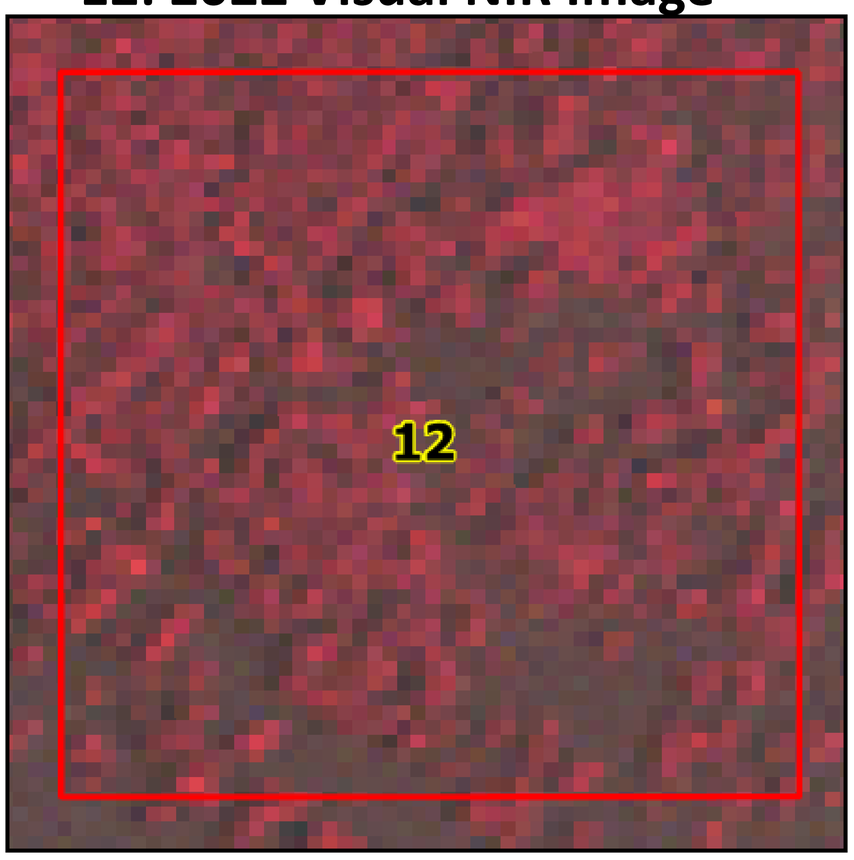
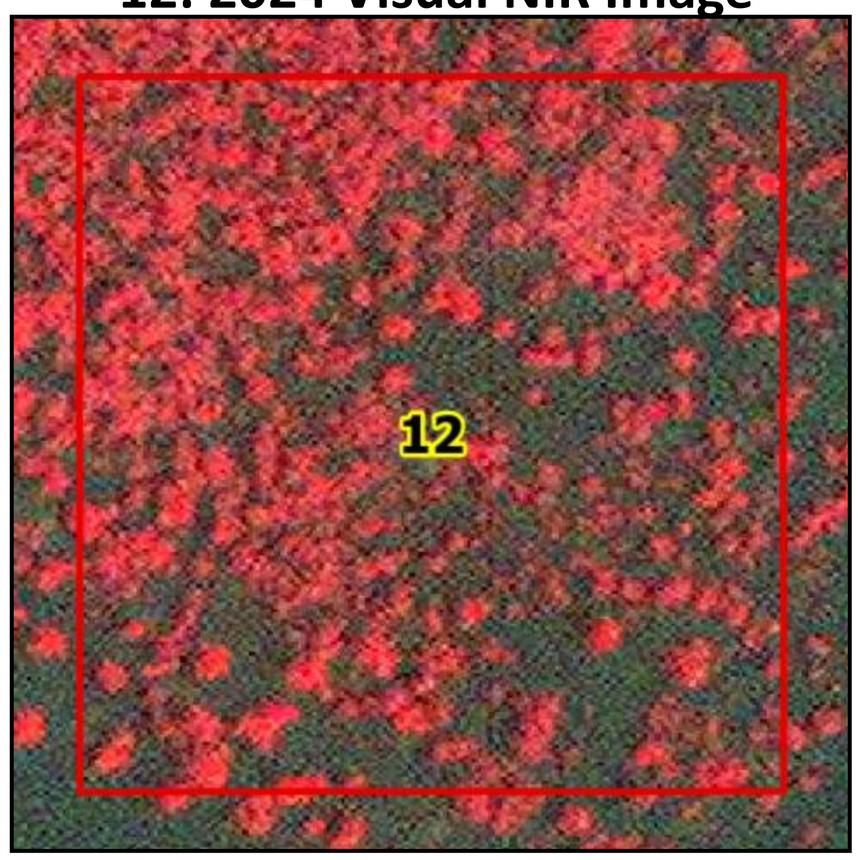
BODO REMEDIATION ANALYTICS-2024 LAND USE/LAND COVER STUDIES

12: 2022 Visual NIR Image

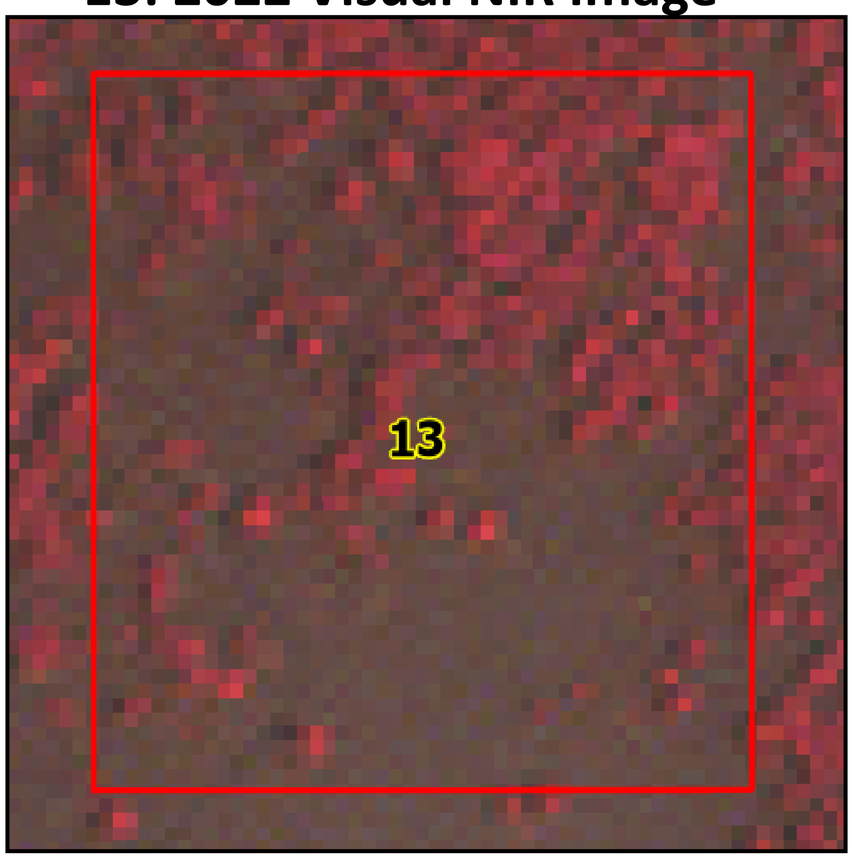


12: 2024 Visual NIR Image

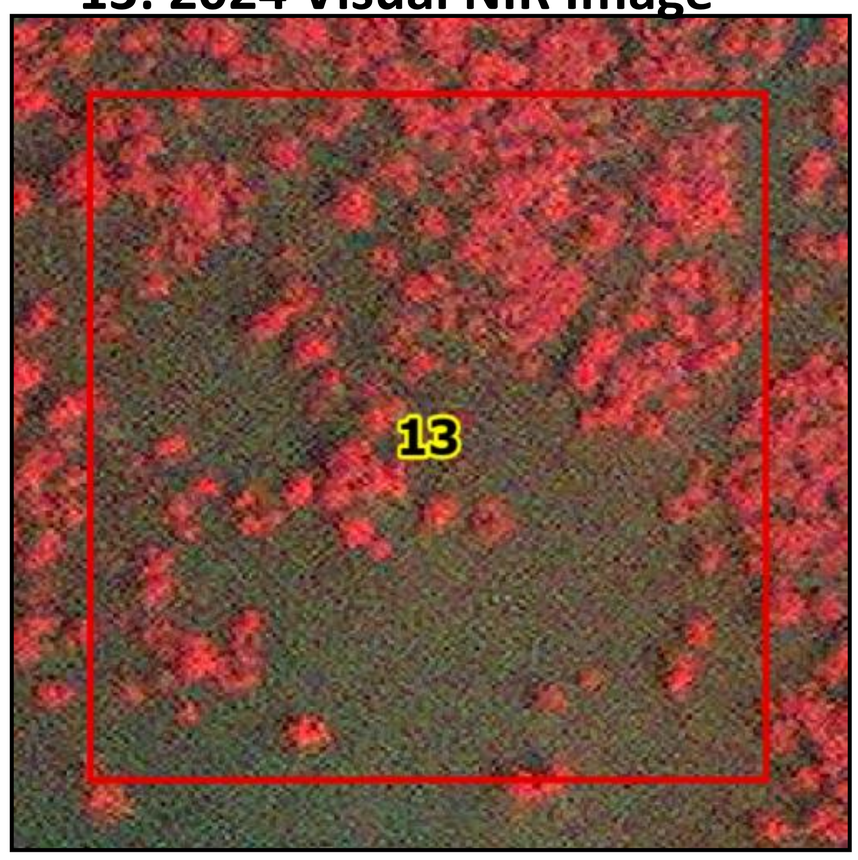


12 - The distribution and pattern of vegetation appear similar between both year of study. There are no significant growth or difference observed. Vegetation appear healthy.

13: 2022 Visual NIR Image

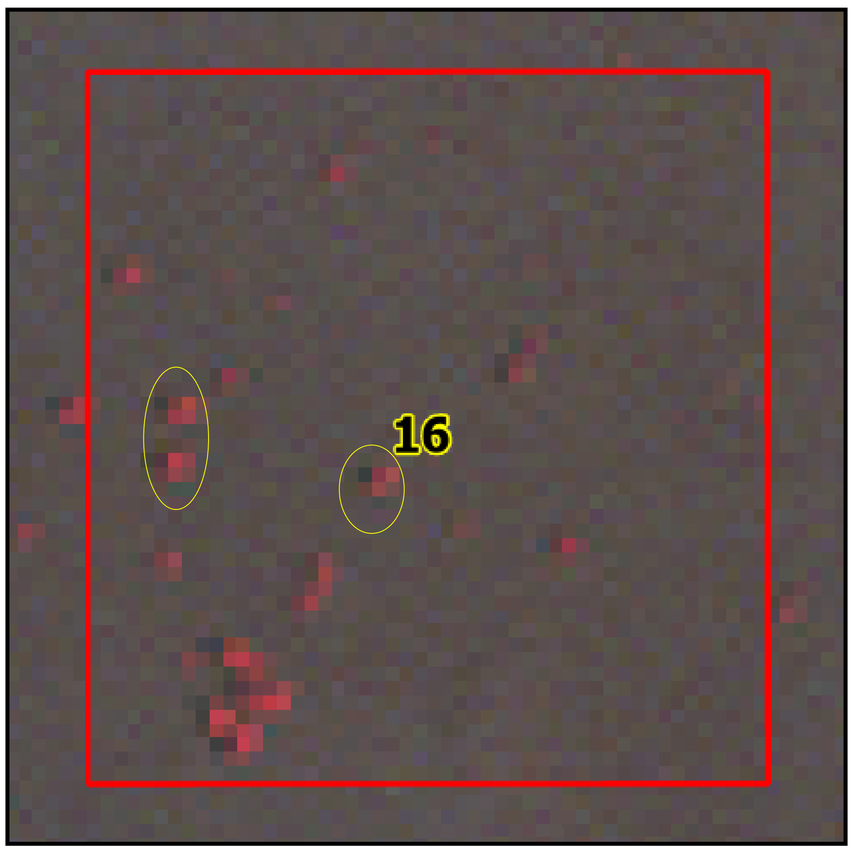


13: 2024 Visual NIR Image

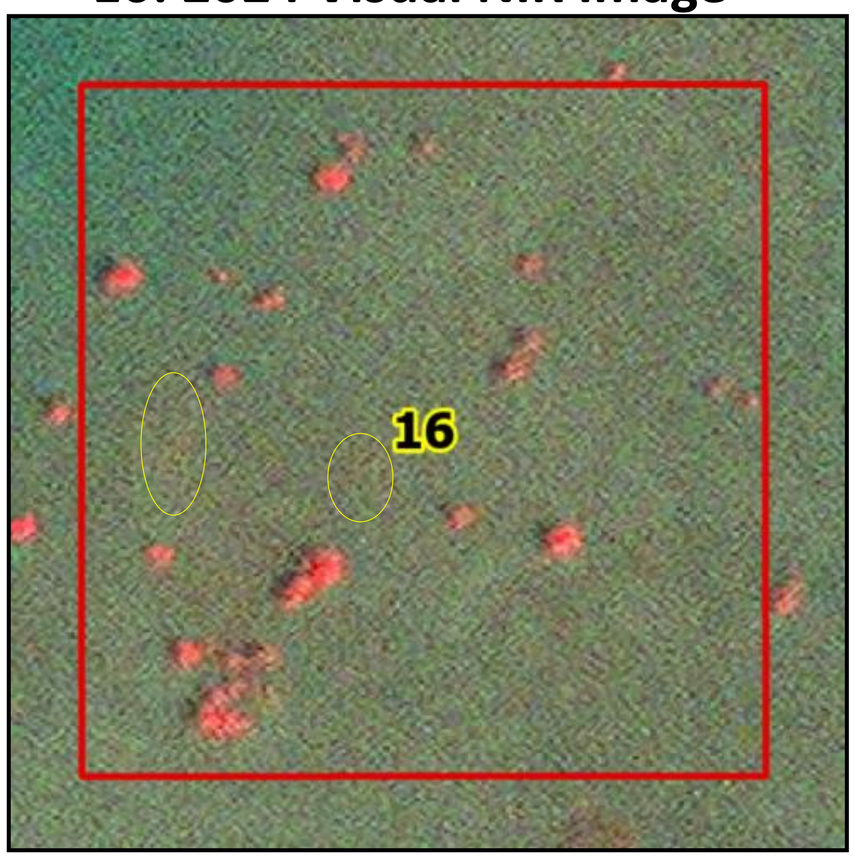


13 - The distribution and pattern of vegetation appear similar between both year of study. There are no significant growth or difference observed. Vegetation appear healthy.

16: 2022 Visual NIR Image



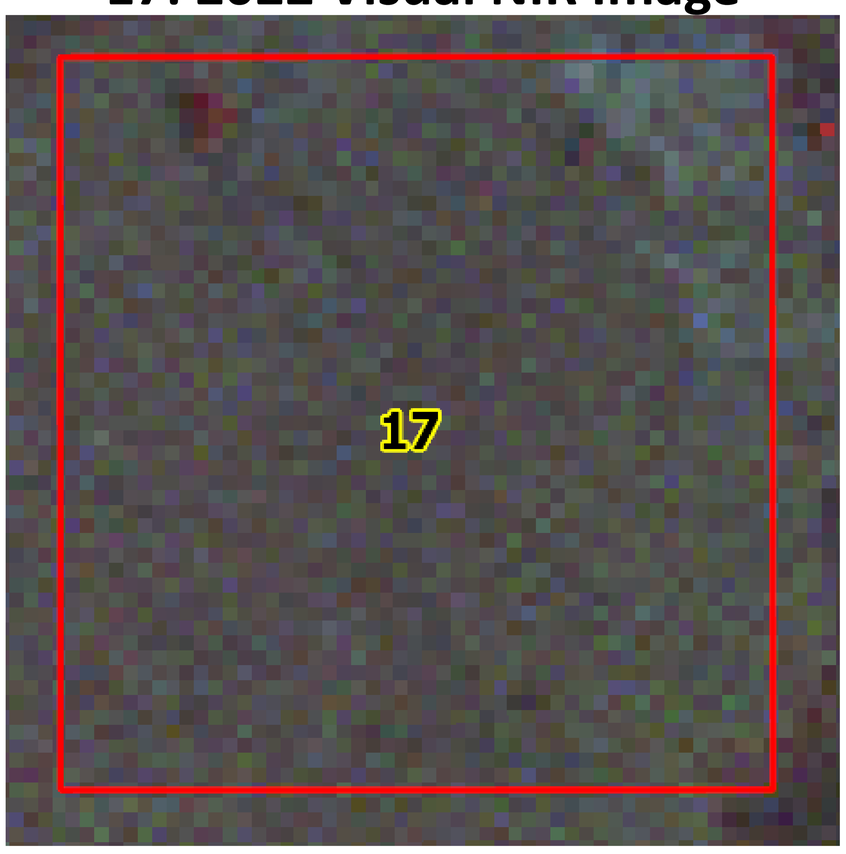
16: 2024 Visual NIR Image



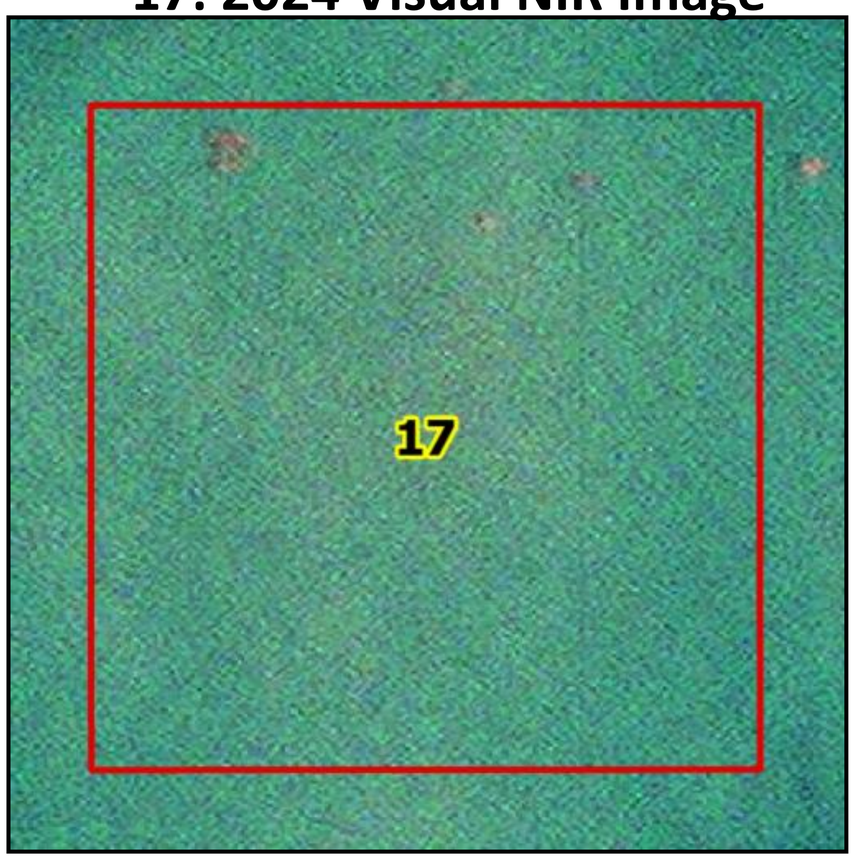
16 - There are no significant growth observed between both images. A noticeable disappearance in vegetation foliage, circled in yellow, was observed in the 2024 image

BODO REMEDIATION ANALYTICS-2024 LAND USE/LAND COVER STUDIES

17: 2022 Visual NIR Image



17: 2024 Visual NIR Image

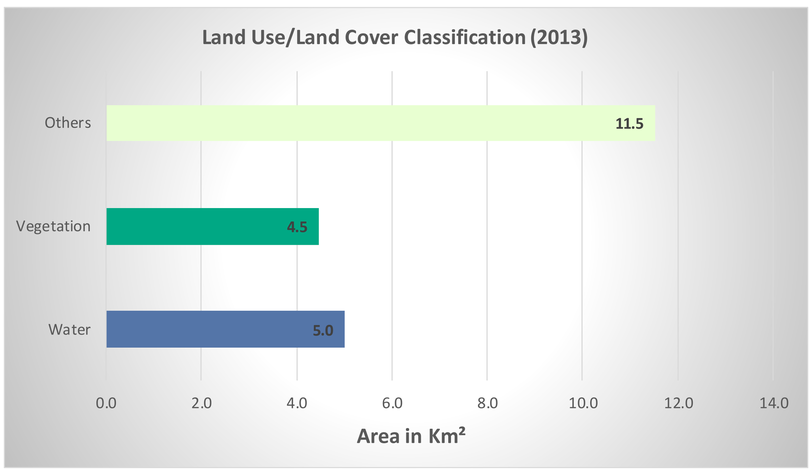


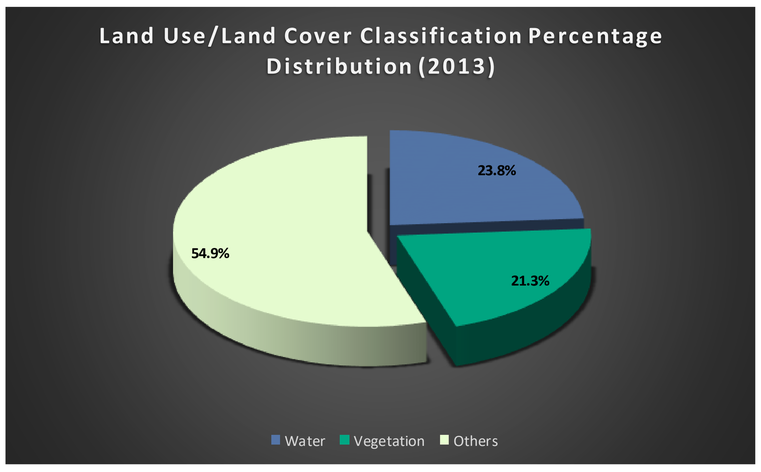
17 - There are no significant growth observed between both images. Cell 17 seems to be devoid of any vegetation regrowth since the last study.

BODO CLASSIFICATION OUTPUT AND STATISTICAL ANALYSIS (2013 – 2024)

2013

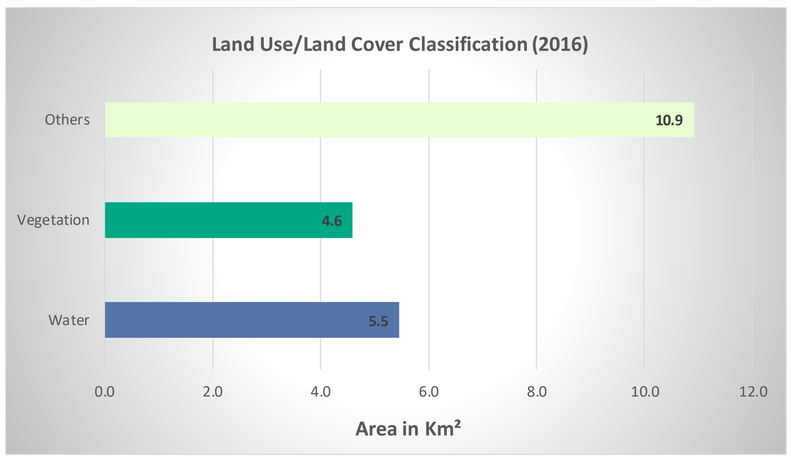
Epoch: 2013		
Land Use/Land Cover Types	**Area (Km2)**	**Percentage (%)**
Water	5.0	23.8
Vegetation	4.5	21.3
Others	11.5	54.9
TOTAL	20.98	100

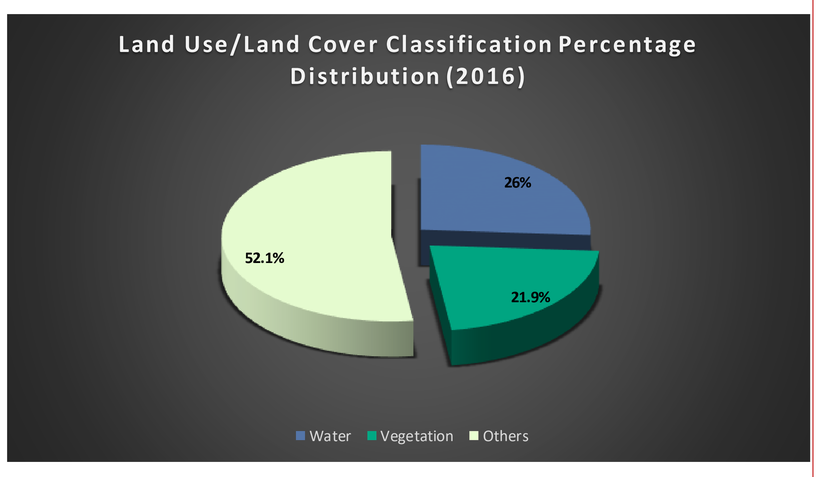




2016

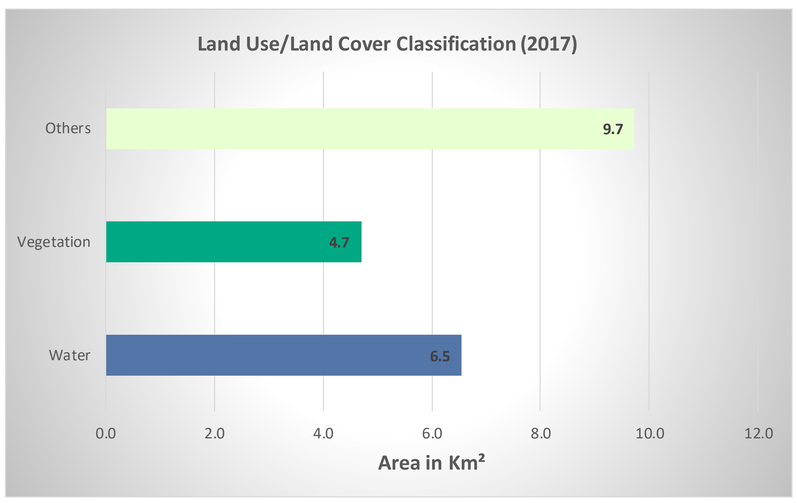
Epoch: 2016		
Land Use/Land Cover Types	**Area (Km2)**	**Percentage (%)**
Water	5.5	26.0
Vegetation	4.6	21.9
Others	10.9	52.1
TOTAL	20.97	100

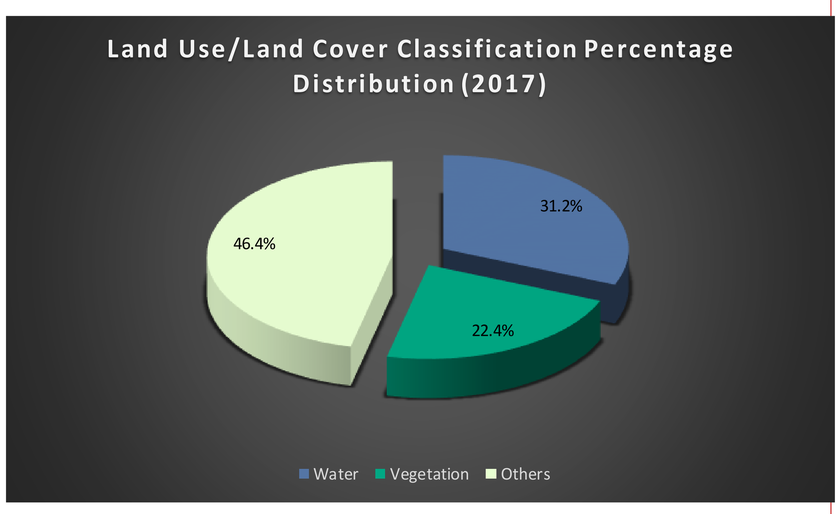




2017

Epoch: 2017		
Land Use/Land Cover Types	**Area (Km2)**	**Percentage (%)**
Water	6.5	31.2
Vegetation	4.7	22.4
Others	9.7	46.4
TOTAL	20.97	100





PREPARED BY: Amaechi Ojeh(IUV/C/NGI)

Geomatics Project Team: Adeoye Ogunleye (IUV/C/NGI), Amaechi Ojeh (IUV/C/NGI), Tunde Shebolatan (UPV/C/NGI)

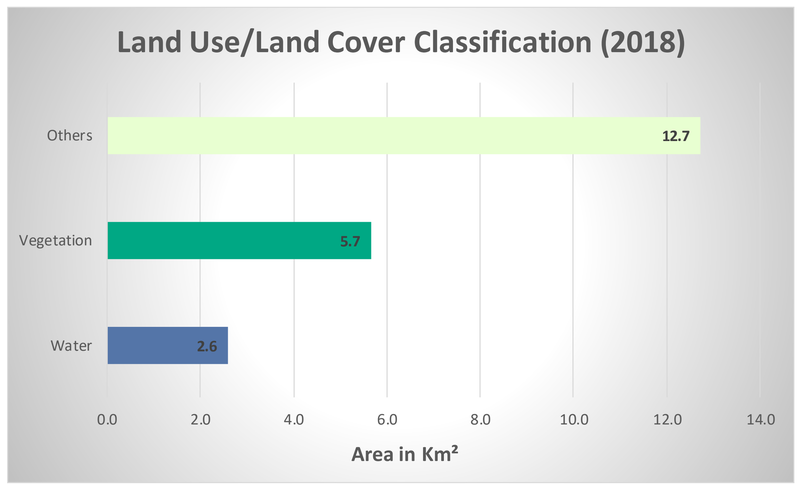
DATE: 23 May 2024

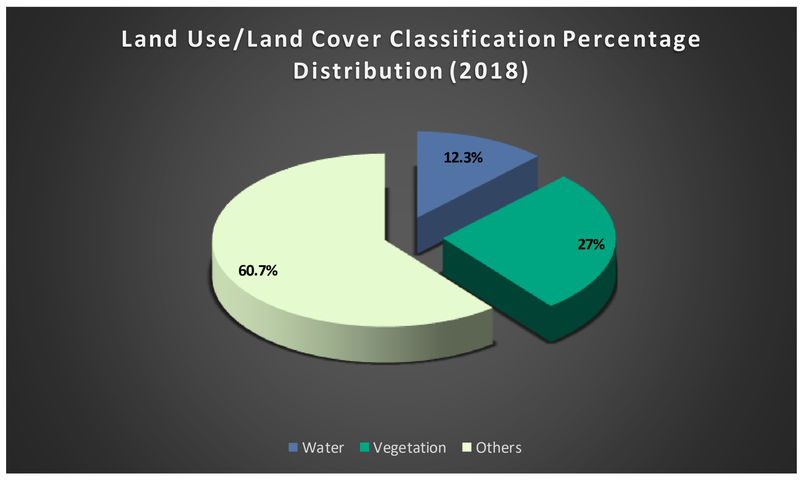
BODO REMEDIATION ANALYTICS-2024 LAND USE/LAND COVER STUDIES

BODO CLASSIFICATION OUTPUT AND STATISTICAL ANALYSIS (2013 – 2024 CONT)

2018

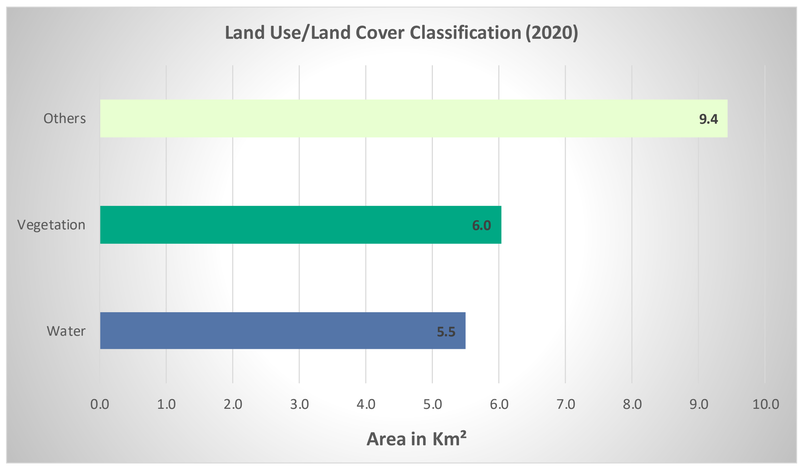
Epoch: 2018		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	2.6	12.3
Vegetation	5.7	27.0
Others	12.7	60.7
TOTAL	20.98	100

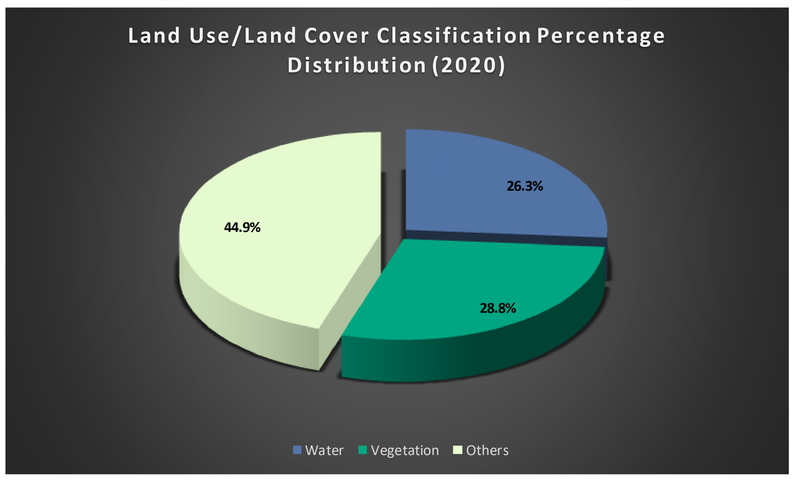




2020

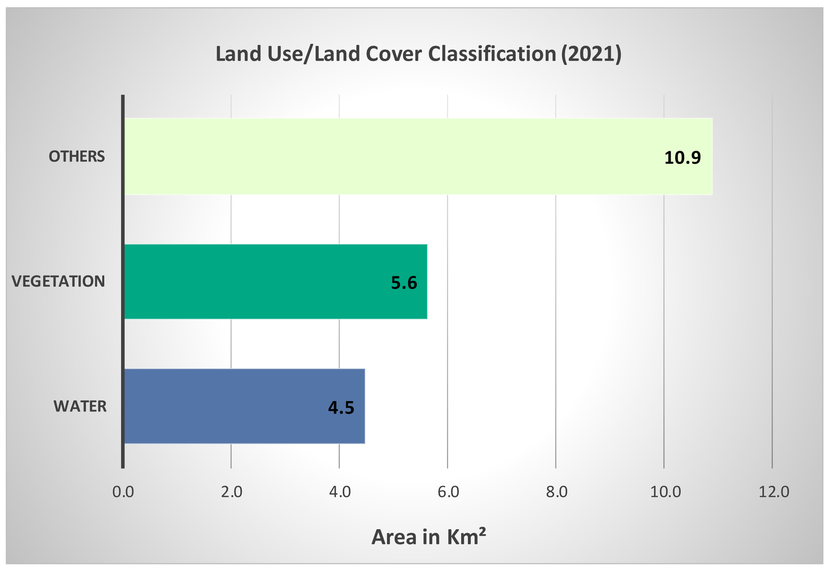
Epoch: 2020		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	5.5	26.3
Vegetation	6.0	28.8
Others	9.4	44.9
TOTAL	20.98	100

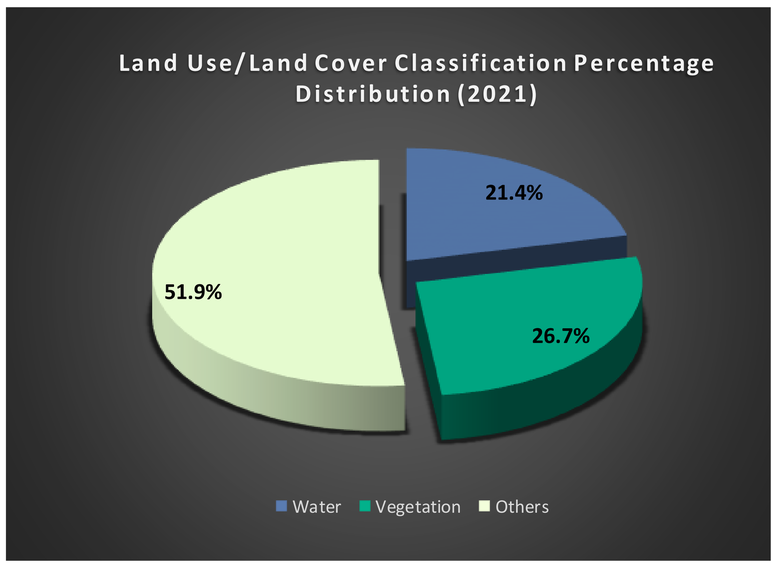




2021

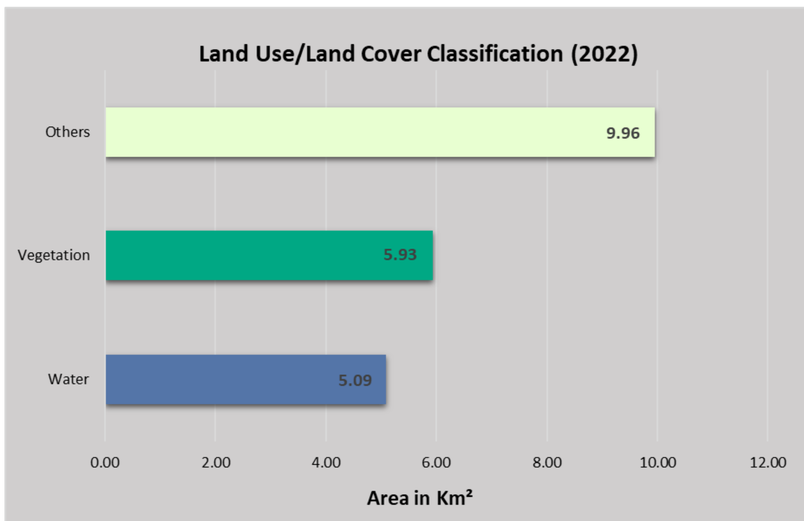
Epoch: 2021		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	4.5	21.4
Vegetation	5.6	26.7
Others	10.9	51.9
TOTAL	20.98	100

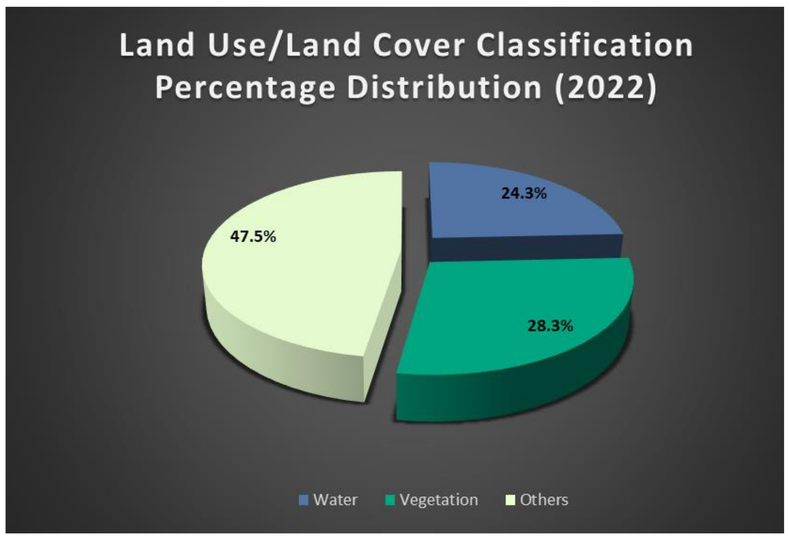




2022

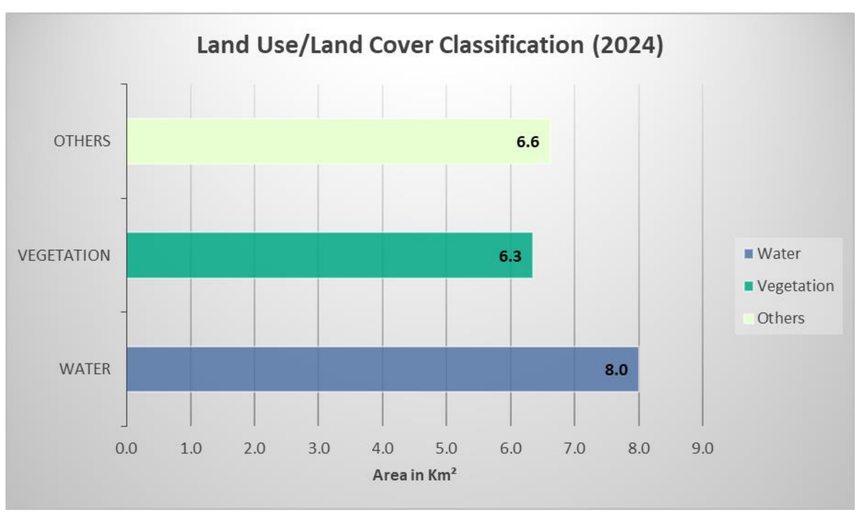
Epoch: 2022		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	5.09	24.3
Vegetation	5.93	28.3
Others	9.96	47.5
TOTAL	20.98	100

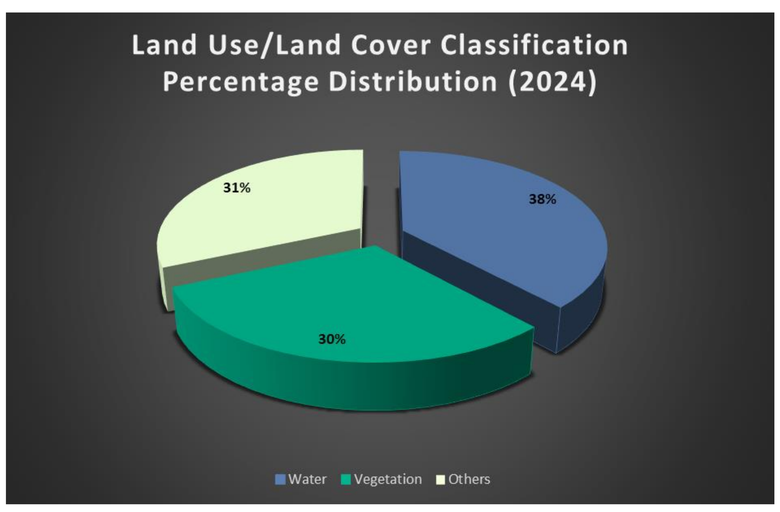




2024

Epoch: 2024		
Land Use/Land Cover Types	Area (Km2)	Percentage (%)
Water	8.0	38%
Vegetation	6.3	30%
Others	6.6	31%
TOTAL	20.98	100%



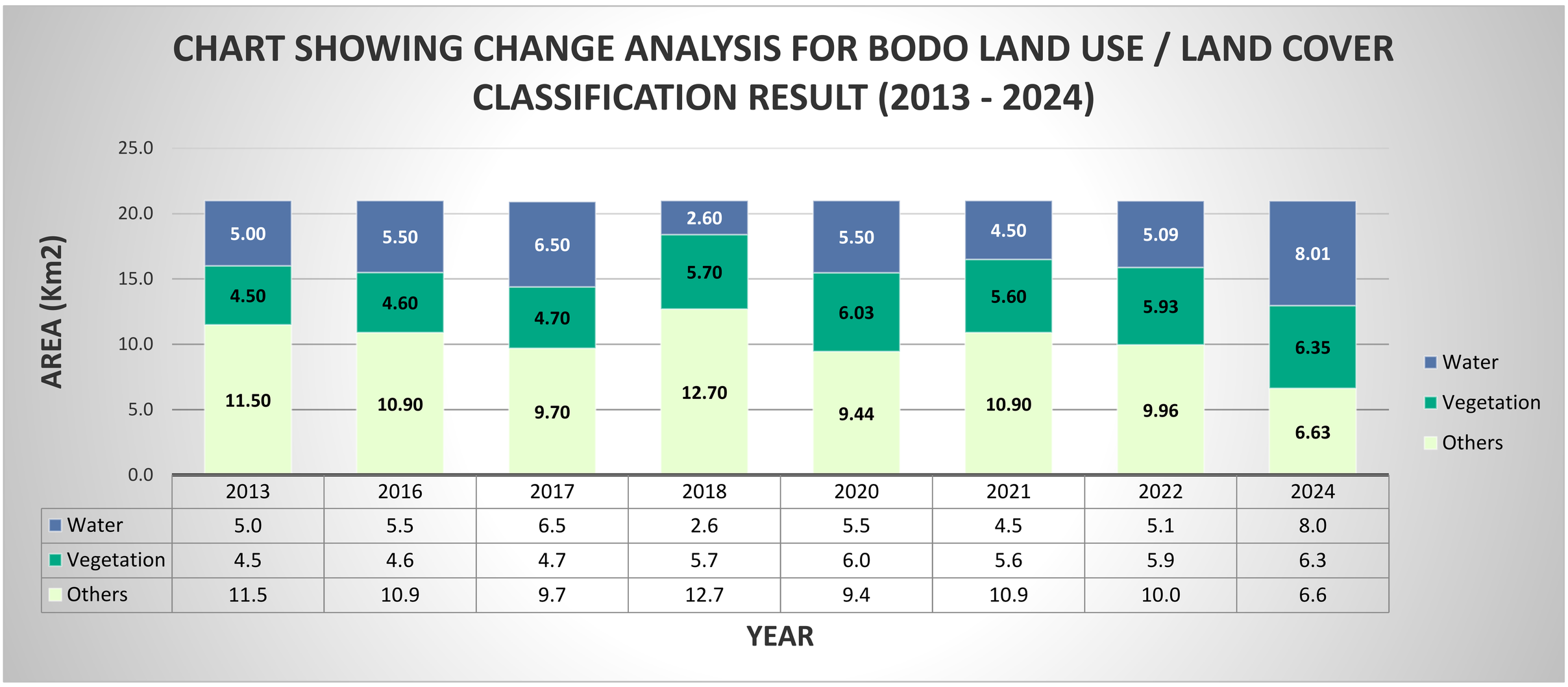


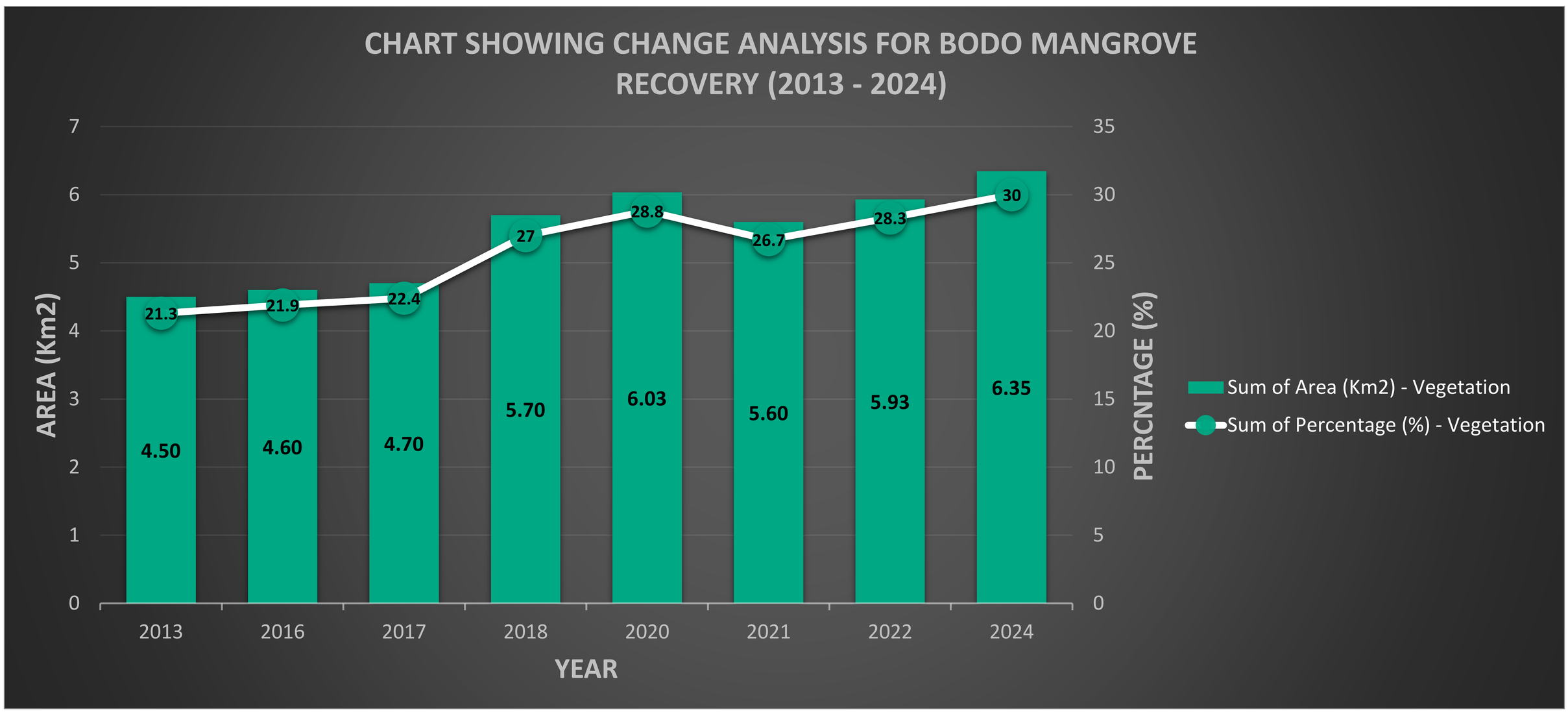
PREPARED BY: Amaechi Ojeh(IUV/C/NGI)

Geomatics Project Team: Adeoye Ogunleye (IUV/C/NGI), Amaechi Ojeh (IUV/C/NGI), Tunde Shebolatan (UPV/C/NGI)

DATE: 23 May 2024

BODO REMEDIATION ANALYTICS-2024 LAND USE/LAND COVER STUDIES

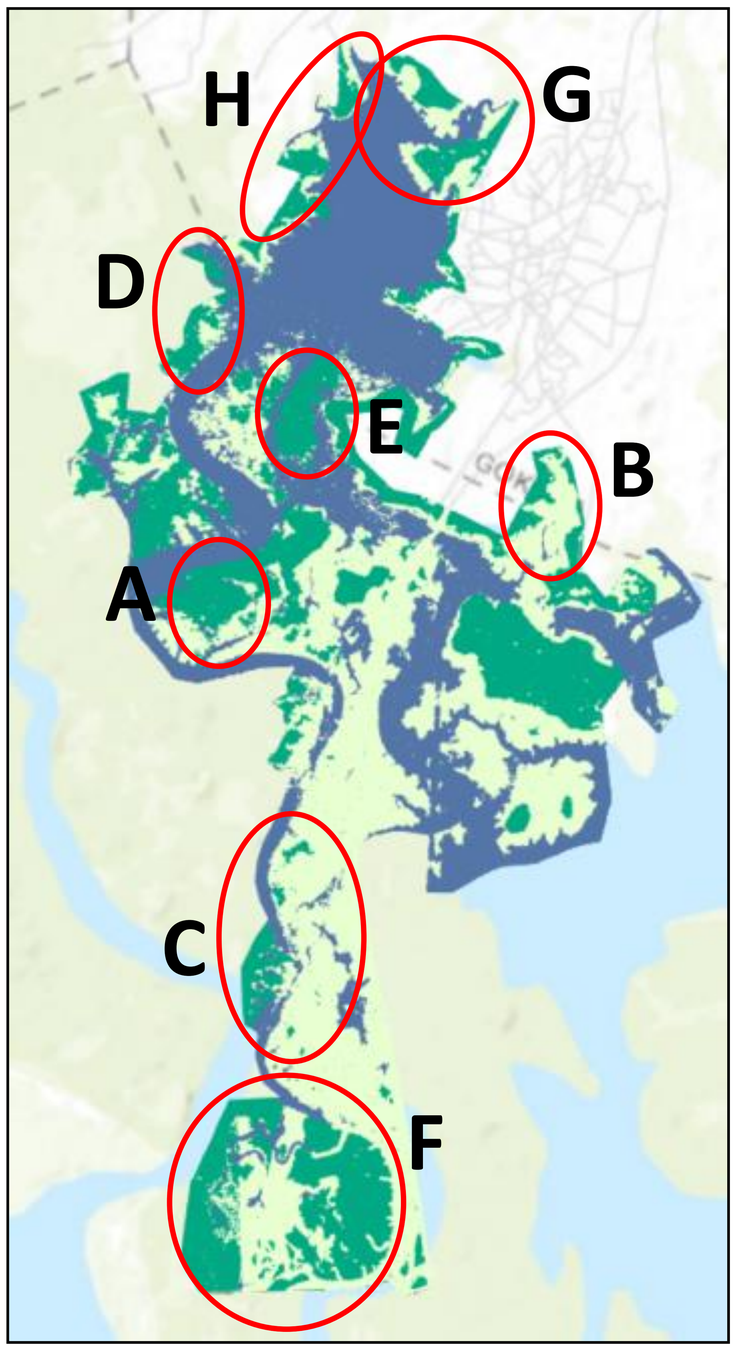




BODO REMEDIATION ANALYTICS-2024 LAND USE/LAND COVER STUDIES

CHANGE ANALYSIS FOR MANGROVE RECOVERY WITHIN BODO AOI

EPOCH 2024



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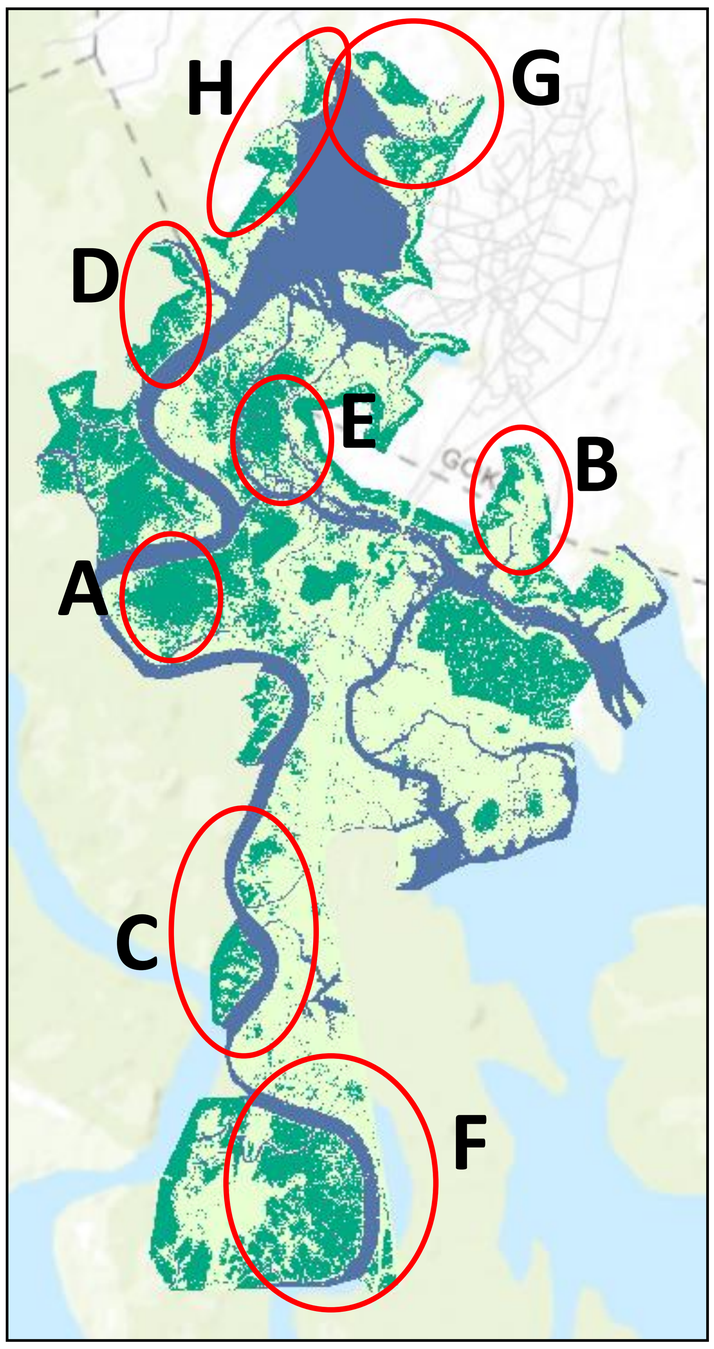

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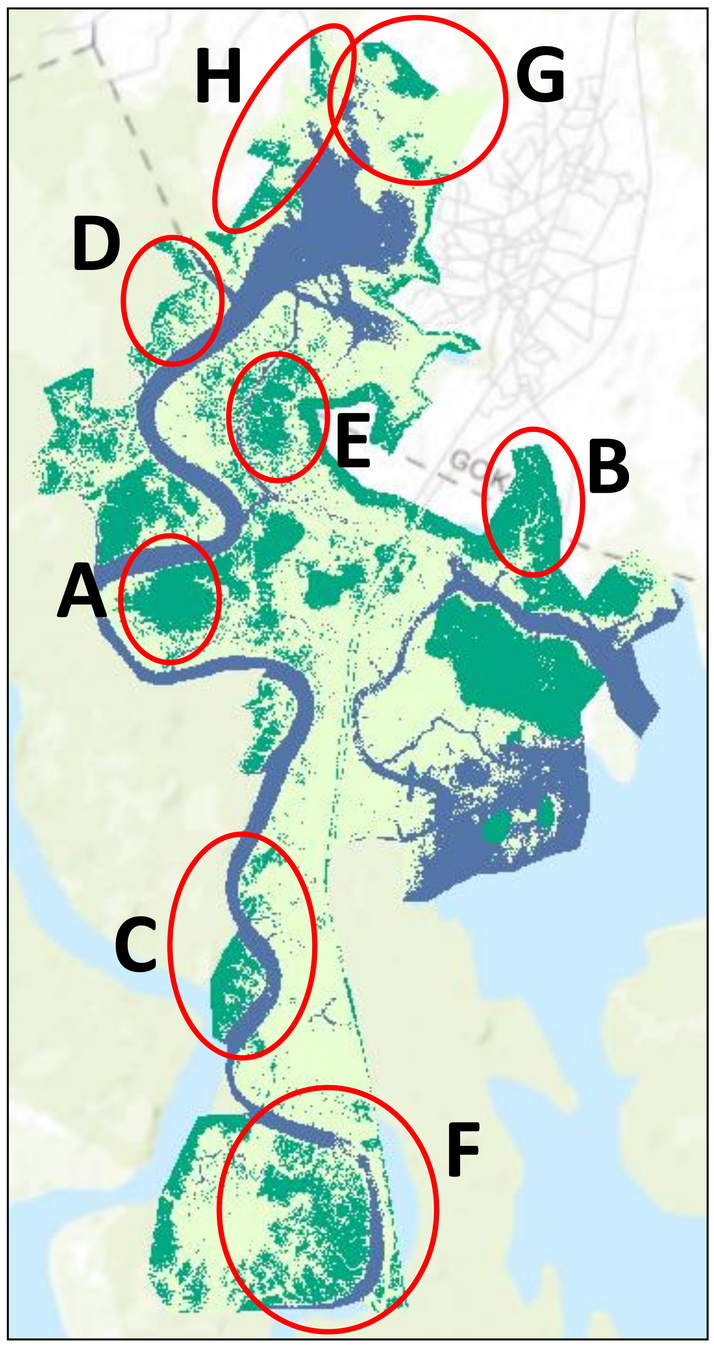

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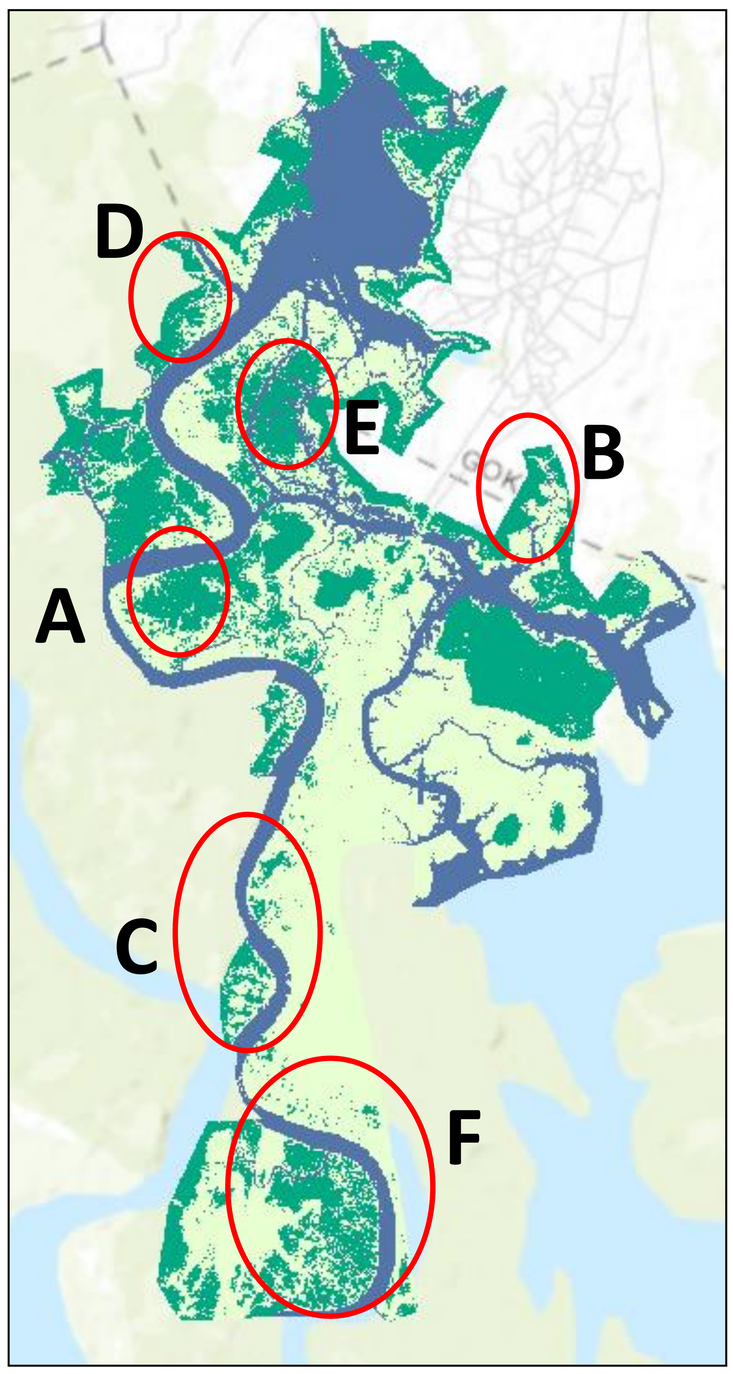

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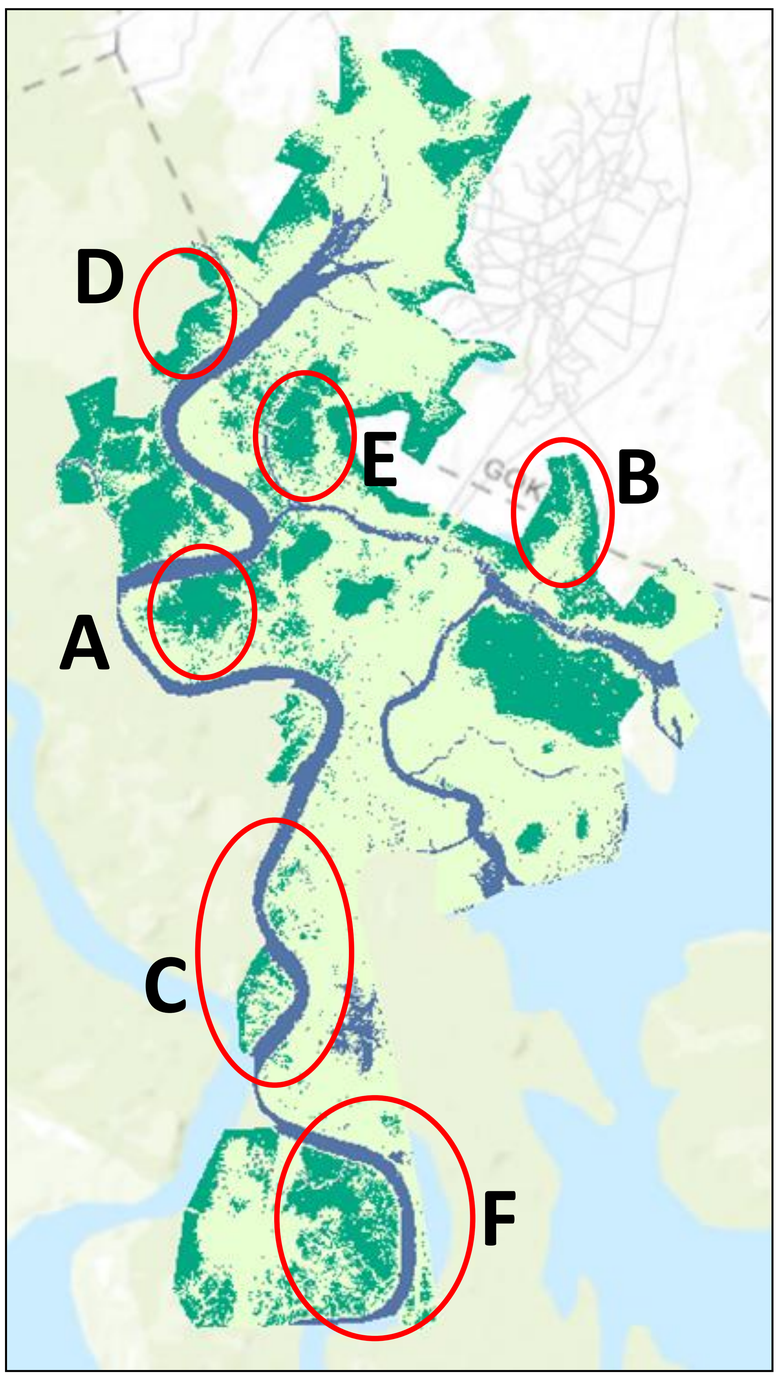

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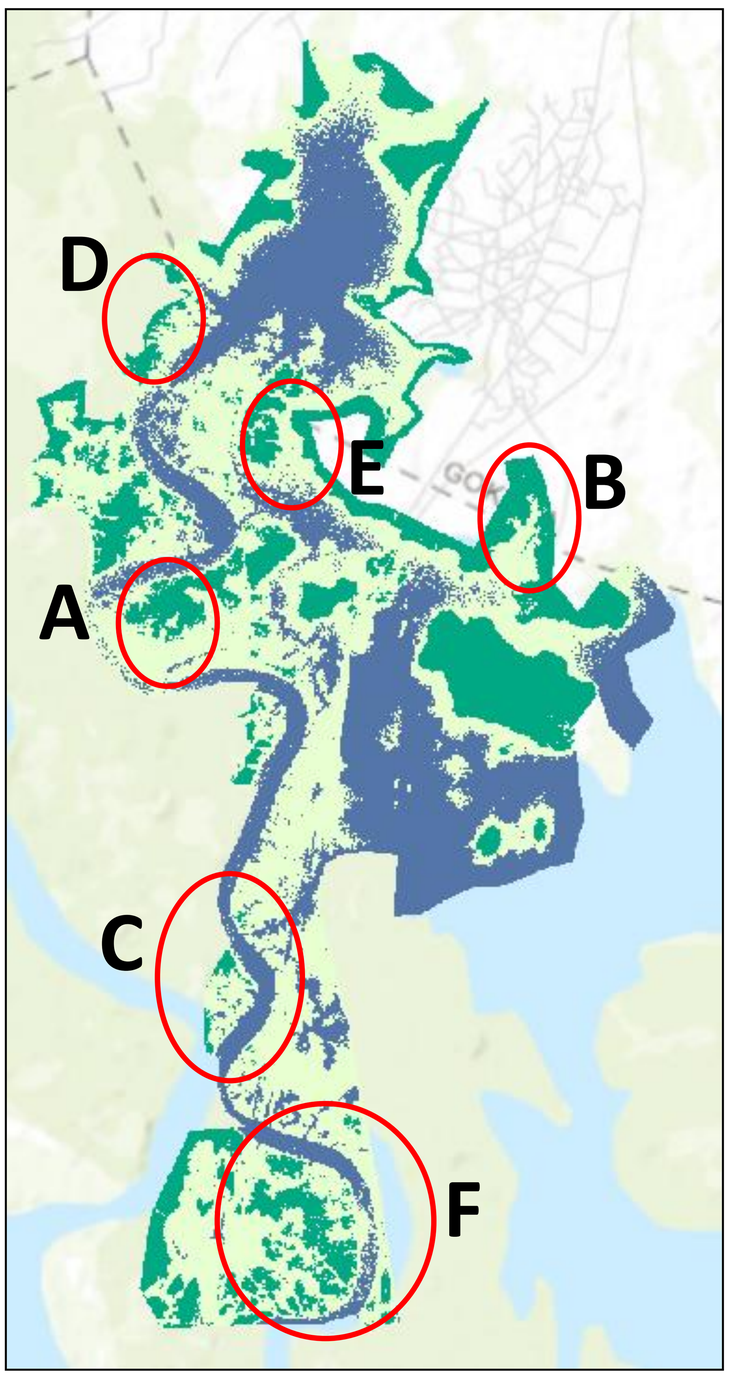


PREPARED BY: Amaechi Ojeh(UPV/C/NGI)

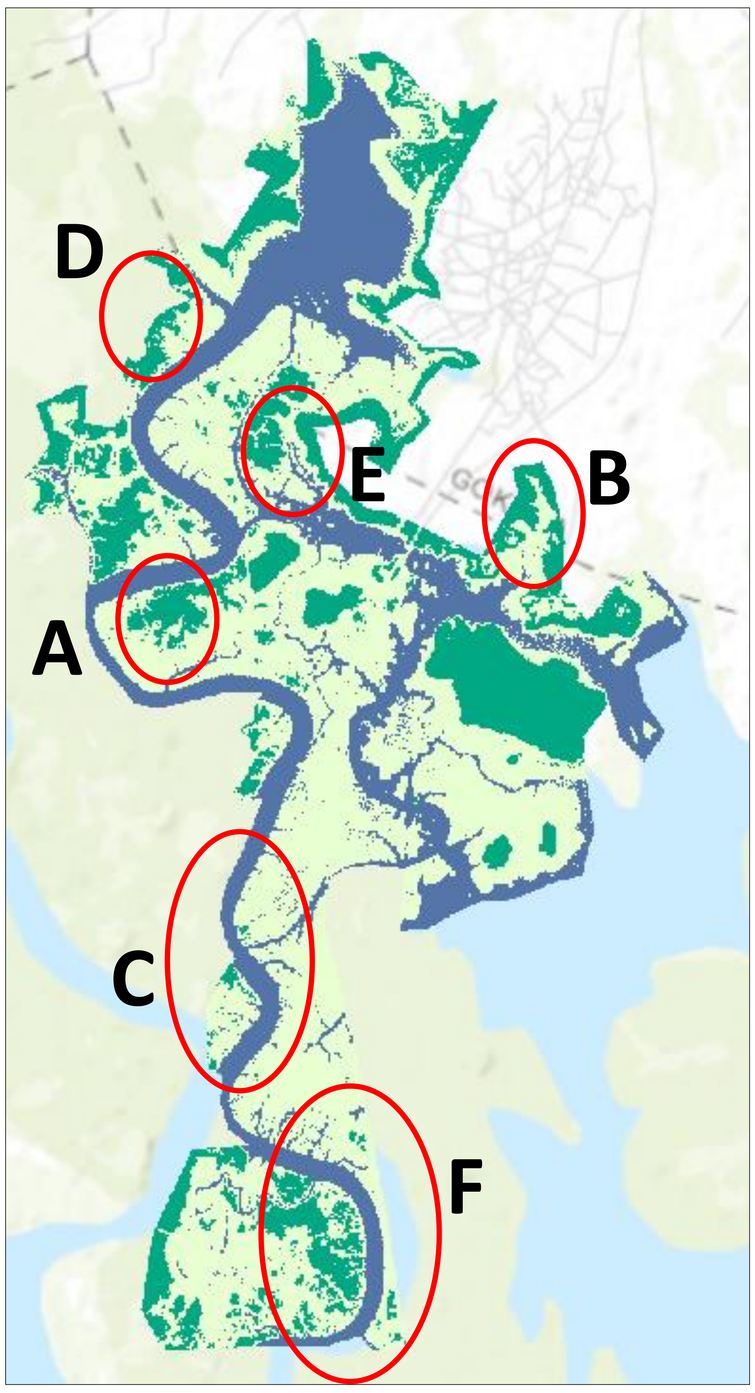
Geomatics Project Team: Simisola Oyetunde (UPV/C/NGI), Amaechi Ojeh (UPV/C/NGI)

DATE: 20th April, 2022

BODO REMEDIATION ANALYTICS-2024 LAND USE/LAND COVER STUDIES

CHANGE ANALYSIS FOR MANGROVE RECOVERY WITHIN BODO AOI

EPOCH 2016



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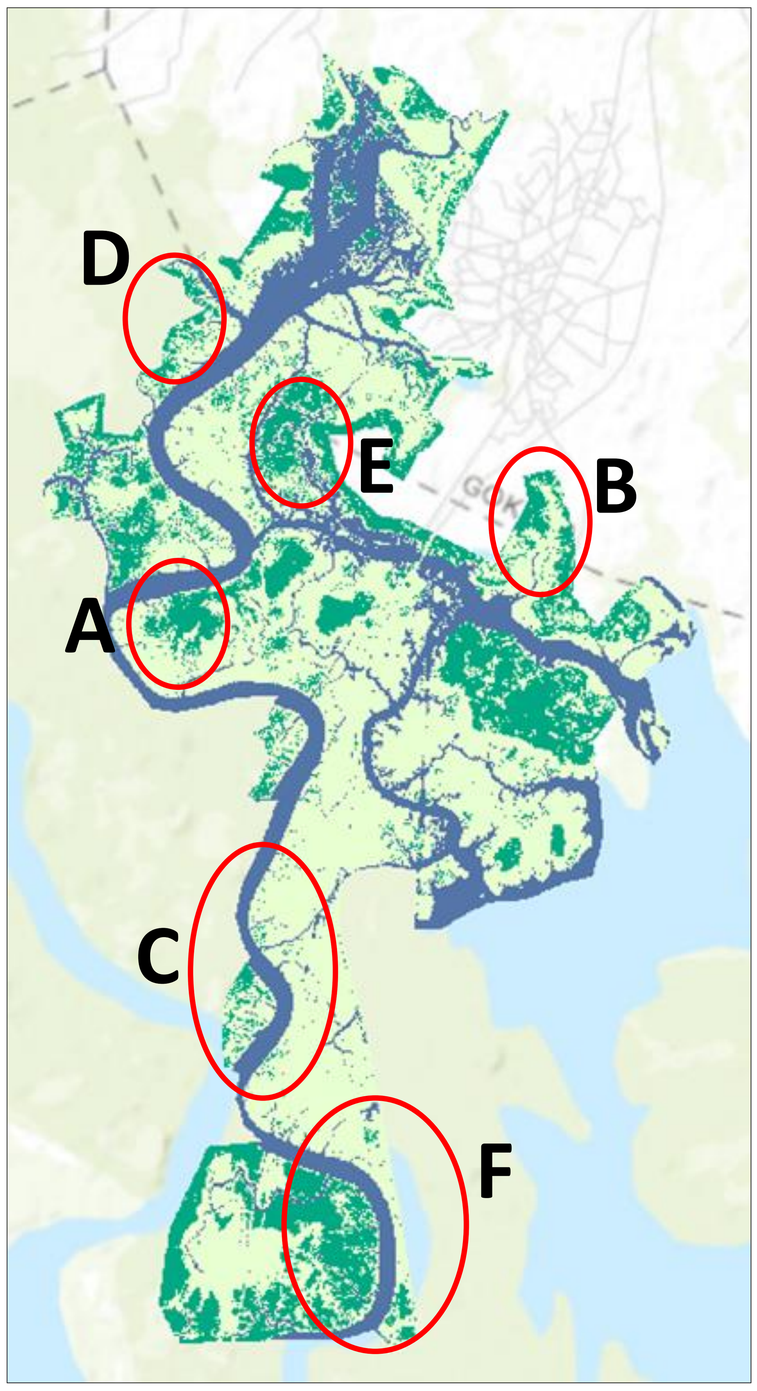

PREPARED BY: Amaechi Ojeh(IUV/C/NGI)

Geomatics Project Team: Adeoye Ogunleye (IUV/C/NGI), Amaechi Ojeh (IUV/C/NGI), Tunde Shebolatan (UPV/C/NGI)

DATE: 23 May 2024

TABLE SHOWING THE AREA AND PERCENTAGE OF THE LAND USE / LAND COVER CLASSES WITHIN BODO AOI FROM 2013 – 2024

Year	Land Use/Land Cover Types	Area (Km2)	Percentage (%)
2013	Water	5.0	23.8
	Vegetation	4.5	21.3
	Others	11.5	54.9
2016	Water	5.5	26.0
	Vegetation	4.6	21.9
	Others	10.9	52.1
2017	Water	6.5	31.2
	Vegetation	4.7	22.4
	Others	9.7	46.4
2018	Water	2.6	12.3
	Vegetation	5.7	27.0
	Others	12.7	60.7
2020	Water	5.5	26.3
	Vegetation	6.0	28.8
	Others	9.4	44.9
2021	Water	4.5	21.4
	Vegetation	5.6	26.7
	Others	10.9	51.9
2022	Water	5.09	24.3
	Vegetation	5.93	28.3
	Others	9.96	47.5
2024	Water	8.01	38.0
	Vegetation	6.35	30.0
	Others	6.63	31.0